

Program overview

19-May-2025 11:19

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master CME 2024			
Master Construction Management and Engineering 2024			
CME Compulsory Courses			
AR8003TU	Legal and Governance	5	
CME1201	Collaborative Design and Engineering	5	
CME2300	Financial Engineering	4	
CME4001	Project Management	5	
CME4121	Information Systems for the Construction Industry	5	
CME4150	Managing Uncertainty and Data	4	
CME4200	Intercultural Relations and Diversity Competence	2	
CME4300	Engineering Asset Management Systems	5	
CME5021	Research and Development Methods for CME	5	
CME Graduation Domains			
CME Systems & Decision Support			
CME4481	Systems Engineering Design	5	
CME4501	Engineering Systems Optimisation	5	
CME4701	Construction Management Systems	5	
CME Must choose at least one, but may also choose multiple			
CME4170	Construction Technology	5	
EPA122A	Spatial Data Science	5	
TPM024A	Methods for Risk Analysis and Management	5	
CME Projects & People			
CME2202	Dynamic Control of Projects	5	
CME4100	Process Management	5	
CME4601	Leadership and Strategic Management	5	
CME Must choose at least one, but may also choose multiple			
CME4800	Procurement of Complex Public Projects	5	
CME4850	Forms of Collaboration in Civil Engineering	5	
TPM024A	Methods for Risk Analysis and Management	5	
CME Design & Integration			
CME4100	Process Management	5	
CME4250	Transformative Design	10	
CME4400	Entrepreneurial Engineering	5	
CME Ethics Courses Must choose at least 1 out of these 4 courses			
TPM003A	Water Ethics	5	
WM0312CIE	Philosophy, Technology Assessment and Ethics for CIE	4	
WM0329TU	Ethics and Engineering	6	
WM0376TU	Ethics of Technological Risks	5	
Writing Skills (optional, but at maximum only one of these courses can be chosen)			
TPM305A	Writing a Masters Thesis in English	2	
WM0201TU	Technical Writing	2	
Project Courses (optional, but at maximum only one of these courses can be chosen)			
CME2100-11	Research Internship	10	
CME4040-10EC	Joint Interdisciplinary Internship	10	
CME4900	Open Design Contractor Project	10	
CME4950	CME Research and Development Project	10	
CME Thesis			
CME5100	CME Master Thesis Preparation	5	
CME5200	CME Master Thesis	30	
CME Thesis for students who started before September 2020			
CME2000	Master Thesis	32	
CME2001	Master Thesis, Preparation	4	

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Introduction 1

For more information, please check the regulations: <http://studenten.tudelft.nl/en/students/faculty-specific/ceg/regulations-rules-and-guidelines/>

Year	2024/2025
Organization	Civil Engineering and Geosciences
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CME Compulsory Courses

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr. F.A.M. Hobma	
Instructor	Prof.mr.dr. E.M. Bruggeman	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and recognize important aspects of the governance modes that develop within the institutional framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of the aspects mentioned above and to find the best possible solutions; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

CME1201	Collaborative Design and Engineering	5
Responsible Instructor	Dr. E.J. Houwing	
Instructor	Y. Liu	
Instructor	M.A. Buijnsters	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, students learn how to collaborate in large multidisciplinary teams on an ill-defined engineering assignment. Each group has to deal with an overload of information that needs to be structured in the design process, in order to identify the underlying engineering problem and present a solution to a client in an effective and efficient way. During the course, students are taught how to identify bad team performance and its effect on the design process, and how to intervene. Students are introduced to the field of Group Dynamics in a series of three workshops (Partnering Charter; Belbin-test; giving and receiving feedback) in the first half of the course. The outcomes of the workshops help students to organize the creation of joint deliverables: Partnering Charter, Project Management Plan, Phase I-report (analysis) and Phase II-report (design-solutions). Overall, students are exposed to an experience of intensive groupwork. Grading is based both on group and individual performance.	
Study Goals	After the course, students have gained an understanding of how to personally navigate and streamline collaborative design processes in a complex environment. Collaborative skills include: knowing how to identify shared values in a large group by making a partnering charter; how to reflect on ones personal role and behavior in a group; and how to give and receive feedback. Moreover, students learn how to acknowledge and successfully deal with different forms of disciplinary knowledge in their group. Students are thus prepared for the realities of practice where engineering assignments tend to ill-defined in the early stages of a project, and multidisciplinary collaboration is a indispensable. The learning objectives for the course are defined as followed: - Discover own team role in the team in such a way that team performance improves considerably, take once a leading role to understand what is going on in the team - Collaborate actively with the team members on various sub-assignments and contribute to the overall design (product) - Describe the main attributes, implement group dynamic theories and apply these for the team configuration in order to be in control throughout the entire assignment - Analyse feedback to personal behaviour in the group and apply new insights in this behaviour in order to improve team performance - Write consistent professional reports	
Education Method	The course is a project course. There are two lectures every week that introduce and give guidance to the project. There is no examination of this course. The assessment is done on the team (group) and individual (personal) results. This project is about a case. The students are divided into numerous (depending on the number of students) teams that each take on the case. The students will divide the roles and tasks within their own team. The first stage of this project is primarily about gathering information and analyzing the interests of different parties of the case. Numerous workshops will guide the students how to order, analyze and present the data. Collaboration and cooperation within the groups are leading for that. The first stage is finalized with a report and a presentation by the team on their analysis. In the second stage of the project, the students will prepare a proposal based on the results of the first stage. Workshops given, will focus on the individual contribution (growth of the individual competences) towards group work. At the end of the course, each team will produce a report and give a final presentation to the client, who will judge the final results and will select the best solution from the group presentations.	
Assessment	The final grade is based on both group and individual assignments (50/50). Assessment is based both on the quality of the joint deliverables and their presentation, online peer-assessment and the self-reflection report.	
Expected prior Knowledge	Positive result for Intercultural Relations. The course is therefore focused on disciplinary knowledge and personality as drivers for group dynamics, rather than cultural diversity.	
Academic Skills	Analytical thinking/ Writing reports/ Cooperation/ Professionalism/ Debating and discussion / Reflection on own personal and group behavior/ T-shaped engineering profile.	
Literature & Study Materials	Reader & Smulders et al. (2012) Teamworks by students, for students.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	No exam is included in this course	
Collegerama	No	

CME2300	Financial Engineering	4
Responsible Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Compulsory course for master CME - Construction Management & Engineering (4TU)	
Summary	Short overview of the theory of the firm Sources of finance, cost of finance, capital markets Cash flow models, decision rules like payback period, C/B ratio, NPV Financial accounting Financing (international) projects, project finance Political and social factors Portfolio management Financial risk management Influence of future trends on sustainability and circularity on finance	
Course Contents	This course deals with the finance issues related to the implementation of civil engineering projects. It introduces economic engineering concepts and finance-related topics such as project financing and financial accounting. This course requires the student to study in detail: # Finance and the firm, covering topics such as sources of finance, cost of finance, financial structures, working capital management and financial accounting # Capital Budgeting Decisions and Risk, covering topics such as Capital Budgeting, Political and Social factors, Portfolio Management and Risk Considerations # Project finance, covering topics such as international capital markets, stakeholder/actors viewpoints and cash flow modelling of projects including characteristics of typical projects like oil wells, open mining, infrastructure and office buildings # Future trend topics for finance, especially implications of sustainability and circularity	
Study Goals	The final attainments of this course that students are able to: # Understand the financing and financial implications of civil engineering projects from both a firm and a project perspective, including perspectives from financial involved actors # Create a cash flow overview from cost engineering information # Apply investment and financing analytical techniques # Assess a project proposal and investment strategies # Apply risk analytical techniques on financial data of a project or proposal # Understand the project life-cycle and its impact on and relationship with project finance # Apply and understand the valuation of (financial) assets and projects under uncertainty. # The ability of students to deal with uncertain political and social factors and financial risks # Understand future financing and business trends with regards to sustainability and circularity	
Education Method	This course will include lectures, case studies and an individual assignment.	
Books	Principles of Corporate Finance, Brealey, Myers and Allen, Eleventh version or later, Global Edition	
Assessment	The final assessment is a written examination (40 multiple choice questions). The use of an calculator is allowed. No cell phones or tablets are allowed. Continued pandemic risks may imply an alternative examination for 2022/23. - The result of the multiple choice exam should be 6.0 or higher.	
Contact	CME Secretary's Office, CiTG room 6.49, mrs. Sandra Schuchmann - Hagman, s.c.m.schuchmann@tudelft.nl tel: +31 (0)15 27 84774	
Expected prior Knowledge	No prior knowledge is required.	
Academic Skills	To prepare for the future interdisciplinary world of work (engineering and finance, projects), to broaden academic skills with respect to optimization and to obtain interdisciplinary competences.	
Literature & Study Materials	Principles of Corporate Finance, Brealey, Myers and Allen, Eleventh version or later, Global Edition Related academic papers	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/.	
Permitted Materials during Exam	* The use of a calculator is allowed, but no mobile phones or tablets / iPads. * No books or dictionaries are allowed	
Collegerama	No	

CME4001	Project Management	5
Responsible Instructor	Dr. M. Leijten	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course focuses on the project management of construction projects. Projects, in all contemporary industry sectors and specifically in construction, are becoming increasingly complex and challenging with the accelerated advances in technology, new business models, forms of collaboration, availability requirements, and an augmented demand to deliver economic value and a competitive advantage. Managing projects in this environment requires strong leadership skills and proficiency in project management knowledge and practice to be able to achieve the project and organizational objectives on time, on budget and to the satisfaction of stakeholders. The course is designed to provide students with scholarly knowledge in the theory and practice of managing construction projects in order to enhance their career options and prepare them to move into management roles by developing their professionalism, versatility and leadership in an environment of constant change. The course presents foundations of project management, as well as more advanced academic analyses on the matter and has been developed primarily to understand the role of project management in construction. It provides an introduction to the methods and tools of project management and how they function in the process of constructing an asset. The course provides students with the knowledge necessary to understand how projects can be managed. The curriculum follows the competencies required to move a project through its various stages. The course provides a thorough exploration of project management processes and tasks. These include: - Project organisation - Management of value - Planning tasks and control strategies - Contracts and procurement methods - Risk analysis and management	
Study Goals	Intended learning outcomes of this course are: - Understand and apply tools and techniques to effectively analyse and manage construction and engineering projects. - Understand the coherence and interrelatedness of technical and human aspects in managing projects and develop strategies to manage these technical and human aspects in coherence. - Develop management strategies to effectively deliver a construction project. - Develop critical and strategic thinking on and insight in the application of project management tools and techniques. - Report effectively and with a solid argumentation on developed project management strategies. - Collaborate effectively in a team. At the end of the course students should meet these project management learning outcomes: - Understand, apply and critically reflect on key-project management tools for estimation of cost and schedules of projects, as well as resource planning to effectively develop an optimised project development and execution plan. - Understand project complexity and understand, apply and critically reflect on techniques for risk analysis and assessment and risk response planning, to optimise the chance of successful project execution. - Understand, evaluate, apply and critically reflect on key-contributors to a project contracting and procurement strategy. - Understand, apply and critically reflect on a project performance monitoring technique (earned value analysis) - Understand, apply and critically reflect on performance control indicators and measures. - Understand and critically reflect on change management and develop an advice on a change request	
Education Method	Live lectures, interactive sessions and course materials offered online via the virtual learning environment, as well as group assignments. Live lectures will be on campus. Recordings from previous years are available. In addition, there are instructional videos on Brightspace (also on Collegerama). There will be no recordings of interactive sessions.	
Assessment	A number of group assignments require a pass to complete the course. There is also an individual digital exam (see Permitted Materials during Exam for further information). The mark for the individual digital exam will be the final mark for the course, but candidates will only pass the course if they have also received a "pass" for the group assignments. The results of the group assignments and the digital exam remain valid after the end of the academic year. The two grading components do not have to be fulfilled in the same academic year; although it is recommended. A sufficient final mark (5.8 or higher) is required to pass the course. General grading rules of Delft University of Technology apply. Each academic year, two exam opportunities are offered. Formative assessment: The feedback students receive on group assignments helps them to keep track of their level of understanding of the matter in preparation of the exam.	
Tags	Project planning / management	
Expected prior Knowledge	There are no mandatory entrance courses. Proficiency in the English language is required.	
Academic Skills	- Collaboration and communication in a group - Academic writing - Building an argumentation - Analysis of complex situations - Proper referencing	
Literature & Study Materials	Required literature will be announced in the virtual learning environment.	
Judgement	All group assignments require a "pass" and the written exam requires a sufficient mark to pass the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	The exam will be closed book. Only writing materials are allowed, no electronic devices. The exam will be in a lecture hall with computers. It is not allowed to use other computers than the ones set up in the exam hall.	
Collegerama	No	

CME4121	Information Systems for the Construction Industry	5
Responsible Instructor	Dr. Ir. R. Kuttanharappel Soman	
Responsible Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Dr.ir. E. Papadonikolaki	
Contact Hours / Week x/x/x/x	2hrs lectures + 4hrs practicals	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course, Information Systems for Construction Industry, educates students about the intricate components and processes of information systems. The course emphasis lies on how these systems are used in management settings to enhance decision-making capabilities This course covers the following key concepts: 1. The influence of Building Information Modeling on 'design for -ty' elements. 2. Semantic Information Modeling (which includes Geometric 3D Computer Modeling.) 3. The enhancement of cost and time management through computational strategies. 4. The role of Information Systems in improving asset maintainability. 5. The use of Computer-Aided Safety Planning. 6. The utilization of Information Systems for facilitating sustainable transitions. 7. The impact of digital transformation on construction firms and projects. This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Project of Interest (PoI) as a connective vehicle enabling the Open Design Learning (ODL) response. The PoI is either being constructed or already been constructed (must have passed the initial phase)	
Study Goals	By the end of the course, the student will be able to LO1. Explain contemporary information systems concepts, principles, and practices relevant to construction projects. LO2. Examine the relationships between information systems concepts themselves and various phases of a construction project, including design, construction, and operation. LO3: Modify project-specific information workflows and integrate distinct information system concepts to facilitate digital transformation in construction projects. LO4: Critically assess the digitalized project information processes and effectively articulate their advantages and disadvantages LO5: Implement digitalization processes across various project stages and illustrate their application using various software solutions LO6: Formulate a digitalization roadmap for digitizing information in various aspects of the construction firms business model.	
Education Method	The course utilizes the Open Design Learning concept (ODLc). Each week, students explore specific concepts and apply these to a self-selected Project of Interest (PoI) (chosen from a pool of projects or from students network) through their original responses and associated open-glass-box computer models. The instructors promote Information System Management concepts by adopting a reflective practitioner approach, utilizing reference books and engaging in student-led dialogues. The learning experience is further enriched by guest lectures from industry professionals. Weekly, students have two hours of concept exploration and dialogue sessions and four hours of supervised work sessions. During these work sessions, students work on their ODL responses under the guidance of the teacher or constructor.	
Assessment	Formative assessment: 1. Online quiz multiple choice quiz with feedback 2. In-class feedback on ODL response during the development Summative Assessment: Open Design Learning (ODL) response: an original enabler demonstrating the personal learning achievements. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen PoI	
Expected prior Knowledge	Same as CME entry requirement. Basic knowledge and experience of BIM, CAD, computer programming or similar is desired.	
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - conclude towards comprehensive open ended solutions.	
Literature & Study Materials	Reference books: - Borrmann et al, "Building Information Modeling" - Sacks et al, "BIM Handbook" Other material TBA.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	Yes	

CME4150	Managing Uncertainty and Data	4
Responsible Instructor	Dr.ir. M. Nogal Macho	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Ir. J.S. Hoving	
Instructor	Dr. Ir. R. Kuttantherappel Soman	
Contact Hours / Week x/x/x/x	4HC, 2PT	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Construction Managers and Engineers have to make decisions on a daily basis. In some cases, the decisions are critical and can cause a project to be successful or to fail. To ensure rational, unbiased decision-making, construction managers or engineers should be able to observe and make sense of different types of data. They should be able to identify the data they need, the most convenient manner of organizing or presenting the data so it can be better understood and the information they can safely derive from the data. The last point is paramount because any dataset is associated with a given amount of uncertainty. In other words, construction management implies making decisions in conditions of uncertainty, thus, identifying and incorporating such uncertainty into the process will result in well-grounded decisions. The course Managing Uncertainty and Data (MUD) will provide students with the knowledge and tools to adequately deal with data and uncertainty to support their decision-making process during their professional development as construction managers and engineers. The topics presented in the MUD course can be classified into data- and uncertainty-related as follows; DATA Data collection: sensing and monitoring Observation theory Model-based data generation Sensitivity analysis UNCERTAINTY Probability theory Risk and reliability Uncertainty quantification	
Study Goals	By the end of this course, the student should be able to: Perform multivariate analysis of data to extract knowledge via data-driven approaches, modelling tools and uncertainty quantification methods. construct relevant probabilistic models, accounting for the dependence between variables and derive consistent conclusions. Analyse risk and reliability concepts in the evaluation of systems, their characteristics and their behaviour to support decision-making under uncertainty. Propose actions (short-term) or plans (long-term) to address specific construction and management-related problems based on an adequate strategy for, and with a holistic vision of, data and uncertainty management. Accept and relate the different backgrounds and prior knowledge of team members and utilize the inherent multidisciplinary to instigate peer learning and optimise team performance.	
Education Method	The course learning objectives are met using the following teaching concepts: Transfer of theory and knowledge through Lectures & Discussions Engagement with theoretical concepts through Practicals & Workshops Self-study through Videos, Quizzes and Notebooks A multidisciplinary team project, including both supervisor and peer review Formative online tests to monitor learning progress This course has been designed to guarantee you can give solutions to the issues that you will face during your professional life as a construction manager or engineer. With this in mind, the course is underpinned by the following three pillars. (i) Onboarding and synergies within a multidisciplinary environment. The CME student population is a melting pot of different cultural and educational backgrounds and is therefore by default multidisciplinary. To be able to collaborate effectively within the MSc CME and specifically within this course, all CME students must (1) be aware of the existing multidisciplinary among their population, (2) understand that this multidisciplinary is a strength that can be utilized and (3) embrace this multidisciplinary as an opportunity to learn from each other. As part of this course, students work together during a multidisciplinary team project on Uncertainty & Data management. The team composition is determined by the lecturers based on student background and prior education, ensuring multidisciplinary in all project teams. Throughout the course, the project teams will participate in several group sessions and activities. To become aware of the existing multidisciplinary in their project teams, the students in each team will meet during the first week of the period to (1) meet and inform each other of their cultural and educational backgrounds, (2) identify each others strengths and weaknesses and based on this (3) organize and assign their roles and tasks in the project team. (ii) Practice-oriented course. Practice-oriented course. During the course, the students will apply the theoretical concepts to solve a realistic challenge. This will help them understand the relevance of the course in real practice, regardless of the final orientation of their professional careers. To give an answer to the challenge, they will have to prioritise information, make sound assumptions and derive conclusions supported by data, in the same manner as they will do during their professional life. The skills and tools learned during this course become essential to building up a solution to the challenge. Also, the students will realise how as they progress in the course, they will be able to better communicate with their team with more specific terminology, a better-organised data presentation and with statements that are supported by the data rather than based on biased perceptions. (iii) Online progressive self-assessment. Throughout the course, students may test their own learning progress by repeatedly taking an online test that consists of randomised problems regarding uncertainty and data management. This allows the students to monitor their knowledge, current level and points of improvement on the different course topics throughout the course period. This also allows the lecturers to provide feedback and monitor the learning of the students. At the end of the course, the students must have reached the pass threshold for the online tests.	
Assessment	To pass the course, students must pass a team project. The students will only be able to submit the team project if each member of his/her team has reached the pass threshold for the online progressive self-assessment. The grading for the course will be based on the team project only. In the team project, the students will give answers to a given	

	challenge. They should demonstrate their learning and development achievements, by explaining the process, decisions, and strategies adopted based on the concepts learned during the course.
	The team project will be assessed based on the course rubric. To pass the course, the grade should be higher than or equal to 6.
Expected prior Knowledge	General understanding of civil engineering systems. BSc level mathematics (calculus, linear algebra).
Academic Skills	Critical thinking Analytical thinking Problem-solving Judgement skills Reasoning Basic computing programming
Literature & Study Materials	Reader Probability and Observation Theory Teunissen PJG, Tiberius CCJM, Verhagen S Ale, B.J.M., 2009. Risk: An Introduction: The Concepts of Risk, Danger and Chance. Available as PDF file and hardcopy on https://www.bol.com/nl/nl/f/risk/33102297/
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4200	Intercultural Relations and Diversity Competence	2
Responsible Instructor	Dr.ir. V.J. Cortes Arevalo	
Responsible for assignments	Dr. C.A. Benitez Avila	
Co-responsible for assignments	Dr.ir. V.J. Cortes Arevalo	
Contact Hours / Week x/x/x/x	Two workshops and four interactive lectures.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Construction, Management, and Engineering projects are complex, innovative, multi-disciplinary, and intercultural projects developed in a dynamic environment and require engineers to competently combine engineering and organisation skills to achieve better the project expected results. This course helps you to look at intercultural relations as a communication process between people that work together and naturally belong to multiple social groups. The course helps to develop a diversity competence to address communication differences before and after they occur. Through role-playing games, class exercises, and a group essay, the students engage with the theory, reflect on their experience, and explore the role of inter and intra-cultural relations to develop a diversity competence that they can later apply in their professional practice.	
Study Goals	Departing from sociological and psychological theories on communication, this course covers themes about (cultural) diversity and inclusiveness in professional collaboration through the following learning objectives. The course takes a dynamic and multifaceted approach to culture as its starting point. At the end of this course, you will be able to... Discover and discuss the cultural and diversity dimensions in social/organisational phenomena, events, and activities related to the construction, management, and engineering practice. Apply the TOPOI model to assess and address the impact of these differences on interpersonal and intergroup interactions. Demonstrate cultural and diversity awareness by developing an appropriate and ethical attitude toward inter and intra-cultural collaboration and communication.	
Education Method	Two intensive half-day small-group training sessions of role-play and/or negotiation games (if Covid conditions allow). The aim is to reflect in the role you played. Blended classroom, that requires and expect you to prepare the book chapters, video clips, literature, and popular articles offered in Brightspace as a weekly learning unit. Interactive lectures with exercises and discussions in small groups that guide and provide input to writing the group essay for the test at the end of the course. Self-study, search and reference to relevant literature will support you in the execution and reporting of TOPOI analysis into a reflection report that students will submit in pairs, following the format we will introduce in week 2. See also 'special information' below.	
Course Relations	This course lays a foundation for the CME1201 Collaborative Design and engineering course; to make students aware of the inter and intra-cultural differences to improve their collaboration practice.	
Books	Edwin Hoffman and Arjan Verdooren. 2018. Diversity Competence. Cultures dont meet; people do. Uitgeverij Coutinho. Isbn 978 90 469 0598 2	
Assessment	We expect students' presence and active participation in the two workshops and the interactive lectures, as these provide input for the group essay. Grading is based on a written test in the form of a group reflective essay that applies the TOPOI analysis introduced in the course book by Hoffman and Verdooren (2018). The individual contribution to the group essay and course activities will also be part of the assessment. In case of failure (grade below 5.8), students can hand in a group addition to the project to be awarded a maximum grade of 6.0. The deadline for the addition is in the retake exam week immediately after the course ends. We will announce in Brightspace the exact instruction for the addition.	
Special Information	If circumstances prevent the game-play, some additional online materials may be offered. The course materials are organised in Brightspace in a blended learning fashion, requiring and expecting the students to prepare for the interactive lectures, which include small group discussions of the study material.	
Expected prior Knowledge	No prior knowledge is expected	
Academic Skills	Students are expected to write a short reflective essay in small groups, the format of which is provided.	
Literature & Study Materials	Video clips and a limited number of scientific articles available through BrightSpace. Course book: Edwin Hoffman and Arjan Verdooren. 2018. Diversity Competence. Cultures dont meet; people do. Uitgeverij Coutinho.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	not relevant	
Collegerama	No	

CME4300	Engineering Asset Management Systems	5
Responsible Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Dr.ir. R. Binnekamp	
Instructor	A. Jagadeesh	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Engineering Asset Management (EAM) is a holistic approach to manage a system/portfolio of engineering assets (i.e., physical objects) that enables a certain quality of services (QoS), overarching and/or encompassing the worlds of Project & construction management and Operations & maintenance management (DBMO). EAM strategically balances and designs these QoS performance levels between: (a) economic user demands; (b) ecologic capital supply (c) isonomic organizational and financial means/support; embedded within a multi-stakeholders / multi-dimensional tripartism system context (open-source). EAM is integratively concerned with efficient and effective control and care of both one-off new engineering asset deliveries (projects management) and continuously safeguarding the desired quality of service levels of existing in-service assets (operations management) over the entire service life of the engineering assets base. These project and operations activities will be integrated into a strategic asset management planning system using multi-criteria decision/design optimization modelling (MCDO, open glass-box). Finally, this course will introduce some of the following important EAM topics (1) systems safety (2) resilience (3)outsourcing: a game of cooperation and (4) BIMSI asset management information. The first alternative view within this course is to provide a integrative viewpoint on EAM where it is viewed as a cyclic socio-eco purpose balancing act (economical, ecological, social). The second alternative viewpoint is the use of a service provider for reflectively practice the integrative projects and service management concepts into a multi-level strategic asset management plan. The third alternative viewpoint, is that the SPoIs EAM systems will be improved and or appraised using the Open Design principles: multi-criteria Open glass-box decision analysis/optimization , Open source systems theory, Open ended systems thinking combined with Open loops management. In this course the following concepts are addressed: 1. Your Service Provider of Interest (SPoI) and the Quality of Service (QoS) systems concept (introduction to strategic engineering asset -, project- and operations management and open design systems theory) 2. Zooming OUT the SPoIs embedding systems context and open-sources stakeholder constraints/ regulations 3. Zooming IN the open socio-eco purpose EAM system, tripartism system modelling and Open loops management (ODL, MI,MII) 4. Multi-criteria Quality of Service Decision Evaluation and Design optimization (MCDA/ MCDO) - The Projects Management and/or Operations Management solution space 5. Strategic Asset Management Planning (SAMP) integrating the Project & Construction Management and the Operations & Maintenance Management (the Project Delivery and the Service Operation Plan,: PDP and SOP) 6. Keeping the SPoI engineering assets Safe - Integral System Safety 7. Special EAM topics (1) Systems Resilience engineering (2)Outsourcing: a Game of Cooperation (3) the BIMSI EAM information concept For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Service Provider of Interest (SPoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The SPoI can a transportation-, water management-, energy infrastructure and/or a real-estate system service provider.	
Study Goals	Students should be able: - To understand and be familiarized with the EAM concepts, principles and practices, by dialoguing these with the constructors, by navigating through the EAM reference books and by engaging to the SPoI. - To apply and relate these modellingEAM concepts, by dialoguing and experiencing these with the SPoI and the reflective practice. - To (re)work this EAM knowledge, by transforming and linking the SPoI-dialogues into new openings/ learning insights. - To demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL response integrating all (computer) models.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen Service Provider of Interest (SPoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the EAM concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of (computer) practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SPoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). --	

	The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude &Conspect /Convey /Convince
Expected prior Knowledge	CE, TPM, ABE or other engineering design related BSc.
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - comprehensively conclude and cogently conspect your open ended solutions.
Literature & Study Materials	Reference books Hastings (2015) Physical Asset Management Graham M. Winch (2009). Managing Construction Projects. Wiley-Blackwell. Nigel Slack (2006). Operations Management (5th edition). Prentice-Hall. Crespo Márquez (2007). The Maintenance Management Framework, Springer. Reference reader (content) Binnekamp; Heukelum; Wolfert ;Zhilyaev (2022) Reader - Open Design Systems Reference scientific articles Kammouh, O ; Nogal, M ; Binnekamp, R ; Wolfert, A.R.M. / Multi-system intervention optimization for interdependent infrastructure. In: Automation in Construction. 2021 ; Vol. 127. pp. 1-11. Zhilyaev, D. ; Binnekamp, R. ; Wolfert, A.R.M. / Best Fit for Common Purpose : A Multi-Stakeholder Design Optimization Methodology for Construction Management . In: Buildings. 2022 ; Vol. 12, No. 5.
Judgement	After handing in the ODL response you will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME5021	Research and Development Methods for CME	5
Responsible Instructor	J. Ninan	
Instructor	Dr. M.L.C. de Bruijne	
Instructor	Dr.ir. G.A. van Nederveen	
Contact Hours / Week x/x/x/x	5 hrs lectures per week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This research methodology course is intended for all CME students. The course will cover a variety of topics, such as: - problem identification; - distinction between research and development projects and processes; - defining R&D objectives; - formulating R&D questions; - setting up an R&D framework; - selecting fit-for-purpose research methods associated with data collection and analysis; - Selecting fit-for-purpose development methods including supporting research methods. A broad palette of R&D methods will be introduced and discussed such as literature review, case studies, mixed methods, qualitative comparative analysis, Q-methodology, development methods for decision support systems, digital construction, etcetera. Furthermore, the course will provide attention to skills and practices such as R&D planning, ethics, and reporting and presenting R&D project.	
Study Goals	LO1: Clearly define and write a report on research and development (R&D) gaps and goal for a scientific project. LO2: Design a scientific research and development project and write a report on that. LO3: Critically evaluate and reflect in a report on the definition and design of a scientific research and development project.	
Education Method	The course will be delivered through weekly lectures and workshops. The lectures and workshops will be delivered by experts on each method.	
Assessment	The course assessment will be conducted via biweekly group assignments and a final individual assignment. Assistance meeting and feedbacks on group assignments	
Expected prior Knowledge	No prior knowledge required.	
Academic Skills	The students will develop various skills for academic research such as conducting a literature review, and critical thinking guided by various methods, along with a focus on the R&D process.	
Literature & Study Materials	Lecture materials will be uploaded in Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Graduation Domains

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Systems & Decision Support

CME4481	Systems Engineering Design	5
Responsible Instructor	Dr.ir. R. Binnekamp	
Instructor	Dr.ir. R. Binnekamp	
Instructor	Dr.ir. G.A. van Nederveen	
Contact Hours / Week x/x/x/x	2hrs lectures + 4hrs practicals	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Systems Engineering Design (SED) is a powerful design approach to get from a semi-unstructured multi-stakeholder set of vague and messy goals and needs to a well-defined set of requirements and complementary socio-technical systems. In this course a best fit for common purpose design methodology to design for complex interconnected systems using multi-objective optimization techniques. Within this approach both the subject preference function model (human) and the physical object behavior model (engineering asset) will be integrated. In other words, to design socio-technical systems one needs a sound and integrative mathematical approach (open glass box modelling) to support participatory decision making on new engineering artefacts (open source participation and open ended system theory). It will be demonstrated that this best fit for common purpose design methodology also can be used for Project management and/or Operations management planning applications. The alternative view within this course is to combine multi-objective (non-)linear systems optimization and the theory preference function modeling in a constructivist design methodology, using mathematical simulation and modelling. This Open Design Systems approach is distinctive from pure engineering optimization methods. Within this approach the main concepts of systems design for tY-s, decision and development are linked with a self chosen System of Interest (SoI): 1. Systems theory and thinking and design as problem (dis)solving; 2. Hard engineering asset systems (technical-physical) vs. soft human management systems (social); 3. A posteriori multi-actor design decision systems; 4. A priori multi-actor design decision systems; 5. Non-linearity in mathematical models for design/decision systems; 6. Best fit for common purpose. For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen System of Interest (SoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The SoI can a transportation-, water management-, energy infrastructure and/or a real-estate system.	
Study Goals	To be familiarized with and understand state-of-the art systems engineering management systems concepts, principles and practices, by (i) dialoguing these with the constructors, (ii) navigating through the system engineering management reference documents and (iii) engaging to a self-chosen real-life System of Interest (SoI). To relate and examine these abstract systems engineering management concepts, by (i) constructing SoI specific (computer) models, (ii) dialoguing with these models and (iii) experiencing these with the SoI and its reflective practice. To rework and transform the SoI specific systems engineering management concepts observations, by (i) transforming and linking the dialogues and experiences into new insights and (ii) developing improvement results where applicable. To form an individual judgement and appraise these new insights/results by means of a conspection between these and the original concepts within the specific context of the SoI. To create an original Open Design Learning (ODL) response that (i) integrates all concepts, (computer) models, new insights and developed results and (ii) demonstrates the internalization process of the aforementioned learning goals and conveys the ODL achievements.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen System of Interest (SoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the SED concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of computer practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). -- The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude &Conspect /Convey /Convince	
Expected prior Knowledge	CE, TPM, ABE or other engineering design related BSc, including (non)linear algebra calculus	
Academic Skills	Skills to: - curiously connect to a committed system of interest; - critically convince others with a compelling response; - clearly convey a communicative response; - creatively convert concepts into a comparative response; - construct a properly compiled open glass box model; - contextualize in a combined and open space; - comprehensively conclude and cogently conspect your open ended solutions.	
Literature & Study	General:	

Materials	Blanchard, B.S. and Fabrycky, W.J., "Systems Engineering and Analysis" Print: 5th edition July 2013. Binnekamp; Heukelum; Wolfert ;Zhilyaev (2022) Reader - Open Design Systems (addendum to this reader). Zhilyaev, D. ; Binnekamp, R. ; Wolfert, A.R.M. / Best Fit for Common Purpose : A Multi-Stakeholder Design Optimization Methodology for Construction Management . In: Buildings. 2022 ; Vol. 12, No. 5. Extra: Ackoff, R.A., "Ackoffs Best". Wiley, 1999. Barzilai, J. Preference Function Modelling: The Mathematical Foundations of Decision Theory in: Trends in Multiple Criteria Decision Analysis, Springer, 2010. Binnekamp, R., Preference-based Design. IOS Press, 2010. Dyme, C.L. and Little, P., Engineering Design: A Project-Based Introduction, 2nd Edition ISBN-13: 978-0471256878, 2003. Fisher, R. and Ury, W.L., Getting to Yes. Penguin, 1991. Roozenburg, N.F.M. and Eekels, J., Product Design: Fundamentals and Methods, ISBN-13: 978-0471954651, 1995. Van Gunsteren, L.A., "Quality in Design and Execution of Engineering Practice" ISBN978-1-61499-251-6 (print) 978-1-61499-252-3 (online), 2013.
Judgement	After handing in the ODL response your will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4501	Engineering Systems Optimisation	5
Responsible Instructor	Dr.ir. M. Nogal Macho	
Instructor	T.R. van Woudenberg	
Contact Hours / Week x/x/x/x	2hrs lectures and 2hrs practicals	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Design and process optimization is inherent to Construction Manager's and Engineers' reasoning and decisions taking, for instance, when maximizing effectiveness, minimizing risks, optimizing life circles, etc. Whereas the goal is usually clear, how to achieve such a goal is not so obvious. In fact, the currently common sequential design process for engineering systems is likely to lead to suboptimal solutions. Moreover, decision-makers very often have to navigate between several competing optimization goals at the same time they guarantee the fulfilment of current regulations, e.g., minimizing construction costs and time within a challenging environmental framework. The Engineering Systems Optimization (ESO) course will provide future designers and managers of engineering systems with the knowledge and tools required in their professional development to support a quantitative, model-based, and multi-criteria decision-making process. This oriented-to-practice course has been designed to guarantee students can give solutions to the real problems that will face during their professional life. Some of the topics presented in the ESO course are: Unconstraint optimization Linear and quadratic programming Mix-integer linear programming Non-linear programming Multi-objective optimization and Pareto optimality (Meta)heuristics	
Study Goals	By the end of this course, you should be able to: Translate an engineering-related problem as a goal-oriented mathematical optimization model, through a 4-step process (i.e., problem, model, method, and solution). Formulate a mathematical optimization model that represents an engineering-related problem with meaningful criteria for the selection of variables, objective function, and constraints, considering the implications of the search method and optimal solution, applied to a realistic Case Study. Evaluate the quality of different exact and (meta)heuristic methods to provide a valid solution for the optimization model in different contexts (i.e., research-oriented goals versus daily-practice goals), with consideration of the trade-off between computational time and quality of the solution, applied to a realistic Case Study using a programming platform (e.g., Python or Matlab). Develop solutions to support multiple criteria decision-making in the optimization process through multi-objective optimization. Elaborate strategies for the optimal design and/or management of real engineering systems based on a holistic perspective of the engineering system optimization (i.e., capabilities and limitations of methods and models) and the trade-off between the level of abstraction of the model and the trust in the obtained solution.	
Education Method	This course is designed to adapt the course content to your needs and interests rather than the other way around. With this in mind, the course is divided into two types of sessions; (i) Lectures and Q&A sessions are devoted to the theoretical aspects of the course, where the teacher will present the theoretical ESO background. Based on this initial information and the suggested reading material, you should connect the new concepts to your previous knowledge. To further internalize the new concepts, during the second hour of this session, you, the other students, and the teacher will discuss and give solutions to questions related to applied theory during a Q&A session. The selected collection of questions will include students questions that can be uploaded in Blackboard before the following theoretical session. This activity aims to guarantee your engagement and maximize your reflection on the raised questions. The Lectures and Q&A session is an interactive live session, and your participation is highly recommended. (ii) Practical sessions are devoted to the application of the concepts in which, guided by the teacher, you have to apply the new concepts to your Case Study. This is a live session where you will receive tips, suggestions, and individual feedback from your teachers. Thus, you have to choose a realistic Case Study at the beginning of the course. For these sessions, make sure that you have Python or Matlab and all the optimization packages installed on your computer.	
Assessment	This course is designed to build students knowledge and skills through a continuous learning process. To guarantee an adequate learning process, a formative assessment is introduced within this course. Formative assessment implies that the student will receive regular formal feedback from the teachers, although they won't be graded. Therefore, the students will have to upload four deliverables related to the different topics discussed in the sessions. They will receive the teachers feedback before the next practical session. The students will only be able to submit their final project if 3 out of 4 deliverables have a positive pass The course evaluation will be based on a Final Project. In this Final Project, the student will answer the optimization problem related to the chosen Case Study. The student should demonstrate the learning and development achievements, by explaining the process, decisions, and strategies adopted based on the concepts learned during the course.	
Expected prior Knowledge	Programming skills are recommendable, though non-necessary.	
Academic Skills	Critical thinking Analytical thinking Problem-solving Judgement skills Reasoning Computing programming	
Literature & Study Materials	The following reference material will be used: Castillo, E., Conejo, A. J., Pedregal, P., Garcia, R., & Alguacil, N. (2011). Building and solving mathematical programming models in engineering and science (Vol. 62). John Wiley & Sons. Chang, K. H. (2014). Design theory and methods using CAD/CAE: The computer-aided engineering design series. Academic Press. Dede, T., Kripka, M., Togan, V., Yepes, V., & Rao, R. V. (2019). Usage of Optimization Techniques in Civil Engineering During the Last Two Decades. Curr. Trends Civ. Struct. Eng, 2, 1-17. Scipy Documentation Optimization and root finding. https://docs.scipy.org/doc/scipy/reference/optimize.html#module-scipy.optimize	

	van Woudenberg, T. R., & van Der Meer, F. P. (2021). A grouping method for optimization of steel skeletal structures by applying a combinatorial search algorithm based on a fully stressed design. <i>Engineering Structures</i> , 249, 113299. https://www.dwavesys.com/learn/featured-applications/
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

CME4701	Construction Management Systems	5
Responsible Instructor	Dr.ir. R. Binnekamp	
Contact Hours / Week x/x/x/x	2HC, 4PT	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The Contractor Management Systems (CMS) course is about the way a contractor organizations manages one-off construction projects of infrastructure/real estate systems or delivers ongoing operations and maintenance quality of services, as outsourced contracts from services providers or other clients. Contractors strive to manage their construction projects/service operation organization effectively and efficiently, supported by state of the art management systems. The organization can be triparted into a (1) cultural/ecological purpose, (2) product-service/economic operating model and (3) the organizational/socioeconomic support sub-system. The overall contractors identity is integrated by the interplay of these sub-systems into the so-called socio-eco purpose. The different characteristics of this view can be derived from organizational theory in which the organization is modeled as a living social organism (a living biotope). The organization therefore changes over time as a learning and developing organization. In this she goes through different maturity levels: Pioneer-Expansion-Maturity and/or phases Pioneer-Differentiated-Integrated-Associative phases, including their relevant management & leadership styles. The predominant aspects of construction project management are money, organization, time, risks, information, quality, and safety: i.e., MOTRIQS. For a service operations contractor, these most important drivers are slightly modified into MOSRIQS (where the first S stands for quality of Service and the Q for process Quality). The nature of a construction project management organization is controlling its one-off project goals and deliverables, where the nature of a service operation organization is to care for the engineering assets by safeguarding their quality of services parameters (e.g. reliability, availability, maintainability, safety etc.). Managing a contractor organization means taking different decisions because both project as well as service management are dynamic process on-the-run. For this PFM (preference function modeling) combined MCDA (multi criteria decision analysis) will be used to model and evaluate important managerial decision (e.g. a supplier selection, a construction logistics scenario etc.) This course offers an alternative view on contractor management systems which integrates the management concept of double loop learning, as applied to organizational learning and development, into the management (styles) of construction projects where the contractor organization is seen as an organic living entity fit for socio-eco purpose. Finally, these management systems will be improved and or justified using the Open Design principles: Open glass-box modelling, Open source multi-criteria stakeholders, Open ended systems theory combined with Open loops management. The following concepts are being addressed following the contractors viewpoint: 1. Management Systems and the PMS - Project & Construction Management System (and OMS Operations & Maintenance Service Management system). 2. Construction project planning & mitigation control (and 3C asset QoS optimization) 3. Multi-criteria managerial decision making (MCDA & PFM) 4. Fit for purpose Quality management 5. Information systems the BIMSI concept 6. The developing contractor organization - a living enterprise for the future 7. Associative co-makers and the supply chain For more course info see : https://www.open-design.school/ This course has a constructivist character, in which the student is guided and encouraged to develop open design knowledge & solutions (openings) using the self-chosen Contractor of Interest (CoI) as a connective and practicing vehicle enabling the Open Design Learning (ODL) response. The CoI can a project or service organisation that contracts within transportation-, water management-, energy infrastructure and/or a real-estate system scopes.	
Study Goals	understand and be familiarized with CMS concepts, principles and practices, by dialoguing these with the constructors, by navigating through the CMS reference books and by engaging to the CoI. apply and relate these abstract CMS concepts, by dialoguing and experiencing these with the CoI and reflective practitioners. (re)work this CMS knowledge, by transforming and linking the CoI-dialogues into new insights. demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL response.	
Education Method	The course is based on the Open Design Learning concept (ODLc): every week students are asked to study specific concepts and apply these to a self-chosen Contractor of Interest (CoI) by means of a self-created ODL response and related open-glass-box computer models. The teachers incite the CMS concepts as a reflective practitioner using both reference books and dialogue questions from the students. The students have 2 hours of these concept and dialogue sessions and 4 hours of (computer) practical work sessions per week. During the work sessions, students can work on their ODL response under supervision of the teacher/constructor. During these practicals also Master class sessions will be organized.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a group deliverable based on the self-chosen SPoI (one group delivers one ODL response). The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL response illustrates how the general concepts have been linked and evaluated to the self-chosen SPoI using a: 1) logical review and/or diagramming 2) computer model(s). -- The Open Design Learning commendation principle will be applied as a grading rubric for the ODL response. Both the SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude & Conspect /Convey /Convince	
Expected prior Knowledge		

Academic Skills	CE, TPM, ABE or other engineering design related BSc and preferably CME 4300 engineering asset management course Skills to: curiously connect to a committed system of interest; critically convince others with a compelling response; clearly convey a communicative response; creatively convert concepts into a comparative response; construct a properly compiled open glass box model; contextualize in a combined and open space; comprehensively conclude and cogently conspect your open ended solutions.
Literature & Study Materials	Reference works are L.A. Van Gunsteren (2011) Stakeholder-oriented Project Management: Tools and Concepts F. Glasl. (1997)The Enterprise of the Future: Moral Intuition in Leadership and Organizational Development (Social Ecology) Henry Mintzberg, Structure in fives: designing effective organizations 1992 Chris Argyris and Donald Schon, Organizational Learning II, 1996 Graham Winch, Managing Construction Projects, Wiley Blackwell 2015 Wayne J. Del Pico, Project Control - Integrating Cost and Schedule in Construction, RSMeans 2013
Judgement	Reference papers Kammouh, Omar ; Nogal, Maria ; Binnekamp, Ruud ; Wolfert, A. R.M.Rogier. / Mitigation Controller : Adaptive Simulation Approach for Planning Control Measures in Large Construction Projects. In: Journal of Construction Engineering and Management. 2021 ; Vol. 147, No. 8. Kammouh, Omar ; Kok, M. W.A.(Maurits) ; Nogal, Maria ; Binnekamp, Ruud ; Wolfert, A. R.M.(Rogier). / MitC : Open-source software for construction project control and delay mitigation. In: SoftwareX. 2022 ; Vol. 18. Wolfert, Nederveen , Binnekamp (2022) Fit for purpose Building Information Modelling and Systems Integration (BIMSI) for Better Construction Projects Management, J. of Modern Project Management (JMPM) DOI NUMBER: 1019255/JMPM029011 After handing in the ODL response your will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Must choose at least one, but may also choose multiple

CME4170	Construction Technology	5
Responsible Instructor	Ir. J.P.G. Ramler	
Instructor	Dr.ir. G.A. van Nederveen	
Contact Hours / Week x/x/x/x	6 x 2hrs lectures 6 x 4hrs workshops 1 x 0,5hrs presentation Optionally, an additional 2hr lecture session with award ceremony	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Understanding the nature and implication of selected structural design aspects such as shape, dimensions, material and design approaches on the one hand and the construction considerations such as execution methods, construction logistics (time-phasing) schedules and costs on the other hand and their interdependency in an integrated construction process of a civil engineering structure. This involves thorough knowledge and understanding of project characteristics, control systems, methodology of the process and supporting systems in order to optimise cost driver aspects in conceptual and final execution design. Both the preliminary design and the design for construction will be considered. 6 Lectures on design process, theory and methods, Systems Engineering and technical subjects (e.g. concrete structures, hydraulic structures, construction and maintenance aspects) 6 Working Sessions in which students work on a Case Study of their Project of Interest, supervised by instructors (Design and Projectmanagers from practice). The case study is performed as group work. The case can be selected from a bridge, a tunnel, a jetty or a building. Presentation, as a team, of the group work.	
Study Goals	Upon successful completion of this subject, the student should be able to: 1. To identify the basic elements such as project characteristics, control systems, methodology and supporting systems in an integrated design & construction process for civil engineering structures; 2. To identify characteristics dictating the way a construction project is being managed in practice and emphasis on the construction methods to be adopted when worked out; 3. To apply the basic concepts of Systems Engineering in a civil engineering project; 4. To optimise the process of design and construction in terms of costs, time and execution method in selecting a construction process, a construction schedule and investment in temporary works (eg. cranes, special equipment, formwork, temporary supporting structures); 5. To demonstrate actions which can be taken to control the design process and to assure the quality of the engineering process and the construction process; 6. To generate different design & construction concepts and to select one of them in view of costs, execution time and durability; 7. To implement all these aspects in a real-life case study.	
Education Method	Lectures, instructions, workshops guided by Professionals from practise (Design and Project Managers), case study on a project of interest (bridge, tunnel, jetty or building)	
Assessment	Feedback (team and personal) during workshops Feedback (team) during presentation In between its possible to make appointments for personal/team feedback Feedback (total group) at final lecture with award ceremony Summative: case study (report + presentation)	
Expected prior Knowledge	MSc courses in concrete structures, hydraulic structures, geo-engineering and/or project management	
Academic Skills	Systems Thinking, Evaluate and Validate, Create and Illustrate, Logical Reasoning, Team working.	
Literature & Study Materials	Slides during lectures with references to function modelling and Geo technique.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	Yes	

EPA122A	Spatial Data Science	5
Module Manager	Prof.dr.ir. A. Verbraeck	
Instructor	Dr. G. Marangoni	
Instructor	Dr. T. Chatzivasileiadis	
Co-responsible for assignments	N.Y. Aydin	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Required for	All EPA Master Students	
Expected prior knowledge	Students should have some prior programming experience. It will be beneficial if you have dealt with a functional programming language like R or Python before. If you have never programmed before, I recommend doing a crash course in Python through Coursera or other online services before joining the class. This course is not about learning how to program. If you are not from the TPM faculty and still want to join the course, send the module manager an email with your motivation to join the course before the start of the course. Without receiving such a motivation email, we will not be able to accommodate you fully.	
Course Contents	This class will train students to gather, merge and clean data from multiple sources. We will focus on urban areas with a spatial lens to gain valuable insights into the reality of multiple societal problems like climate change, energy transition, and inequalities. With unprecedented growth in cities, policymakers strive to balance providing equal opportunity and benefit to its citizens and sustainable development. Data science will help us understand and estimate multiple implications of solutions and communicate results to a broad audience effectively. The course is divided into five major modules, each focusing on crucial steps in the lifecycle of a data science project. Obtain : Obtaining data from multiple [open] data sources. Scrub : Data cleaning, munging, and sampling to consolidate all information into a dataset that is manageable, informative and relates to your problem. Explore : Exploratory data analysis to make sense of what your data is trying to say. Model : Estimation and modelling based on statistical tools such as regression, classification and clustering. Interpret : Communicating results and reflections through visualisation, storytelling and interpretable summaries. Students will need their laptops only during labs. All classroom sessions will be a mixture of interactive lectures, discussions and labs. Labs will cater to a series of exercises to reinforce the skills and topics presented in class. A final project will allow students to run through the entire cycle of 5 modules by obtaining and synthesising data to explore a problem of choice and interpreting and communicating their results through an article. For students who have had statistical, math or computer programming courses in their bachelor's or elsewhere, this course will add to your skills by providing you with tools to become future policy-makers, academics, responsible data scientists, and supporters of open science. The course meets 8 hours a week, divided into two interactive and blended lecture hours, 4 lab hours and two extra wildcard hours weekly. Although attendance for all lectures is not mandatory at all, substantial self-study will be required to keep up with the course if you do not attend at least 4 hours a week.	
Study Goals	After successful completion of this course, you will be able to: interpret and discuss data sources usable and relatable for a problem presented. manipulate data and consolidate all information into a manageable, informative dataset that relates to your problem. describe and analyse the consolidated dataset(s) to support your problem with evidence. apply statistical and machine learning models to infer results in turning data into valuable information. report results and reflections through visualisation, storytelling and interpretable summaries, especially when faced with a new dataset. and hopefully, get great data-driven policy jobs in the future, or go on adventurous travels with an open and critical mind.	
Education Method	Out of class education methods involve 1. Reading materials and watching videos assigned before lectures, making notes and asking important questions in class. (Blended Learning) 2. Working on notebooks in Python (ungraded, and useful to stay on track). In class education methods involve 1. Interactive lectures and discussions 2. Labs / open hours to solve assignments	
Computer Use	The course is set in Python programming language. Students will be expected to use Python notebooks on their personal laptops. If this is not possible for you for any reason, please contact me before the course and we will try to figure an alternative solution while maintaining your privacy.	
Books	No books needed	
Reader	No reader needed	
Assessment	0% Individual assessment methods include Exercise (Formative) 50% Individual assessment methods include Assignment (Summative) 50% Group assessment methods include (size: 4 students/group) Project (Summative) It is mandatory to score ≥ 5.8 in both parts separately to pass the course. Partial grades are valid. Only if students get < 5.8 in any part, they may retake that part next year in Q2. For assignments this means retaking all assignments, and for project, this means redoing the whole project with a new group.	

	No exam
Exam Hours	No exam
Elective	Yes
Tags	Adventurous Artificial intelligence Building & Spatial Development Group work Modelling Policy Analysis Programming Programming Software

TPM024A	Methods for Risk Analysis and Management	5
Module Manager	Dr. M. Yang	
Instructor	Dr. M. Yang	
Responsible for assignments	Dr. M. Yang	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course consists of three blocks following a risk analytical approach: (A) introduction, (B) analysis, (C) assessment, treatment and follow-up. In block A Students will get familiar with the concepts and definition used in safety science and the basic steps of a risk analysis. In block B the course will focus on identification of hazards and qualitative/semi quantitative risk analysis. A selection of methods is explored to get acquainted with its application and limitations. For instance: Fault tree analysis, Event tree analysis, Bowtie diagram, Reliability Block Diagram, Markov Chain Analysis, Bayesian networks, Decision making under uncertainty. In block C the assessment of the outcomes of a risk analysis are put central. This includes acceptance, stakeholder perspectives, decision making and monitoring.	
Study Goals	After taking this course the student will have knowledge about: * the basic concepts of safety science and risk analysis * risk acceptance and decision making about risks After following this course, the student will be able to: * apply structured methods for risk identification, risk assessment, risk evaluation, and risk treatment.	
Education Method	The following methods may be used: Lectures, exercises and workgroup assignments. Students are expected to form groups to work on the projects applying risk-based methods in design or operation of socio-technical systems.	
Literature and Study Materials	The following textbook will be partially used: Engineering Risk Management (Meyer and Reniers, 2016; De Gruyter Publishing) Other required and additional material will be distributed via Brightspace	
Assessment	The primary assessment item of the course is a group project with an expected group size of 5-6 (depends on the class size). Assessment will be carried out on basis of groups at different stages of the project period. It will consist of three assignments, one presentation, and one final report. In total, there are 5 summative assessments. The grade distribution is: assignment 1 (proposal) - 15%, assignment 2 (methodology) -15%, assignment 3 (case study and results) - 15%, assignment 4 (oral presentation) - 25%, and assignment 5 (final report) - 30%. The minimum grade for each assessment item is 6,0. Feedback on the graded assignments can be reviewed by students upon appointment with the teach staff of this course within a week after the release of grades. The students are allowed one re-sit per assignment. It is not allowed to request a re-sit for an assignment for which students have received a pass (6,0 or higher). The re-sit format needs to be discussed with the teaching staff in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit must be taken within the current study year.	
Elective	Yes	
Category	MSc level	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Projects & People

CME2202	Dynamic Control of Projects	5
Responsible Instructor	M.G.C. Bosch-Rekvelde	
Instructor	Dr. D.M. Hall	
Instructor	Dr. Ir. R. Kuttanharappel Soman	
Contact Hours / Week x/x/x/x	4hrs lectures + MOOC	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	After the obligatory courses of project management and process management, this course challenges you to reflect upon dynamic control of projects. How should we organize our projects such that we can better deal with dynamics? Each project (green field, brown field) is an intervention in an existing situation. The aim is to incorporate a project (system) into an existing environment. This environment is increasingly complex, hence influencing the project. A project is a specific and unique endeavor, which needs time for its development with a number of subsequent phases: design, engineering, implementation, etc. In most cases: (1) the state of the environment (politics, stakeholders, regulations, etc.) at the start of a project is totally different from the state of the environment during the project, at the end of the project and certainly during the operation phase during the lifetime of the system, (2) the information and knowledge about the desired system in its environment increases dramatically during the project, which leads to different views and options and (3) the state of the technology to be applied in a project changes during a project. As a consequence, we often develop systems with the ideas of yesterday and with today's technology for tomorrow's people. And that's where dynamic control comes into place: we need to be able to act dynamically: the only certainty we have is that uncertainty will hit our projects! To create value, we should not only work dynamically in an adaptive way with continuous change, but also take care that the systems we deliver are adaptable to changing circumstances. Inevitably, the traditional project management should realize that the construction world will also be a part of the new short-cyclic and circular economy. We need to play with complexity; allow for more dynamic approaches in project management and to strive for solutions that maximize value. We need to reorganize our project organization.	
Study Goals	After this course you are able to: 1. Explain the dynamic character of a project, a product and a process, 2. Perform a complexity assessment on a real life complex project, 3. Recognise the relations between complex project systems and higher scale (context) systems, 4. Apply methods to deal with the effectiveness and efficiency of projects in a fast changing world, including novel project production systems 5. Choose management approaches to deal with different types of project complexity (fit for purpose), 6. Describe the advantages and disadvantages of innovative ways of project organising in complex environments.	
Education Method	The concept of "dynamic control" will be elaborated in two lectures a week. Part of the course has an online character (4 lectures). The dynamic practice of projects is illustrated by 4 guest lectures (45 minutes each). Students will write an essay (in small groups) and actively provide peer feedback to other teams in the development of the essay.	
Assessment	The assessment consists of three parts: Essay: to be written in small groups (3 students) Online: two individual assignments, testing the knowledge on the concepts of complexity & management strategy. Game: two or three games should be attended and the related assignment should be submitted. Three partial grades will be given that count as follows: Essay: 2.5 ECTS Online: 1.5 ECTS Game: 1 ECTS Minimum partial grades should be 5, total grade to pass the course should be >5.75. Submissions will be checked for plagiarism. All assignments of the course in Brightspace (ungraded as such) should be completed before the course grade can be obtained. Formative assessment: Students will get feedback on the different submissions steps of the essay (theme & structure literature draft for peer review draft to discuss with staff), see slides lecture 1.	
Expected prior Knowledge	Given the reflective character of this course, you should have successfully completed the CME project and process management courses. Contact the lecturer if this is not the case.	
Academic Skills	Your academic writing skills will be developed by the writing of the essay. Further, reflection skills are essential for successfully completing this course.	
Literature & Study Materials	This course uses the book "Projects & People, Mastering Success", edited by Hans Bakker and Jaap de Kleijn. ISBN 978-90-812162-9-6. Next to the book, academic papers are used (through Brightspace). The MOOC MEP201 on onlinecourses.tudelft.nl is part of this course CME2201.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CME4100	Process Management	5
Responsible Instructor	Dr. M. Leijten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Complex construction- and engineering-related processes require collaboration between a variety of actors within a complex and dynamic environment. An environment that is influenced by various social, political and technical factors. These actors may be related to a process in various ways, e.g. as decision-makers, regulators, risk-taking participants and stakeholders, with diverging interests, resources and attitudes towards the process, being affected by a process and influencing it in different ways and degrees. In addition to the widely used project management, process management is needed to understand and adequately deal with these characteristics. This course aims at providing students with insights, concepts and skills needed to understand the nature of interaction in a multi-actor system regarding the initiation and development of, and decision-making in uncertain and dynamic situations. Students will analyse, develop and apply strategies, tools and arrangements that are part of the process management approach. There will be special attention to transitions and adaptive systems in an infrastructure and construction environment, but the principles are more broadly applicable. There will be a special focus on transition management and adaptive management to address the main current societal challenges related to construction management and infrastructure systems.	
Study Goals	After successfully completing this course, students should: - Be able to identify and analyse the perceptions, interests, strategies and institutional conditions that shape and constrain the behaviour of stakeholders. - Understand the dynamics of multi-actor interaction processes in uncertain and highly politicised contexts. - Understand information exchange in complex processes. - Be able to cope with substantive and strategic uncertainties and to show this in description and application. - Have developed knowledge, skills and competences on how to manage divergent and conflicting interests of stakeholders and the ability to apply this in practice. - Have developed competences on arriving at broadly-shared solutions. - Understand the essentials of decision-making in multi-actor networks. - Understand and apply theories on transition management and adaptive systems. - Have acquired knowledge, skills and competences on arrangements and tools which support collaboration, and be able to apply them.	
Education Method	Teaching is done through live lectures and interactive sessions, as well as learning materials offered via the online learning environment. There is also a group assignment that is offered in the virtual learning environment. Lectures and interactive sessions will be offered either on campus or in the virtual learning environment, depending on possible constraints in effect. Recordings from previous years are available on Collegerama. Interactive sessions are not recorded.	
Assessment	A group assignment (pass/fail) and an individual written exam. Students will receive further information and guidance on the exam during the lecture series and in the virtual learning environment. There will be two exam opportunities every academic year. The mark for the individual written exam will become the final mark for the course, providing the candidate has received a "pass" for the group assignment. Both scores will remain valid after the end of the academic year. A sufficient final mark (5.8 or higher) is required to pass the course. General grading rules of Delft University of Technology apply. The exams will be organised either on campus or remotely, depending on possible constraints in effect.	
Elective	Yes	
Tags	Process	
Expected prior Knowledge	There are no mandatory entrance courses. Proficiency in the English language is necessary.	
Academic Skills	- Collaboration and communication in a group - Academic writing - Building an argumentation - Analysis of complex situations - Proper referencing	
Literature & Study Materials	Required literature will be announced and provided in or via the virtual learning environment.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	In the current set-up, the exam will be closed-book. This means that no materials other than writing materials are allowed. If this policy changes, it will be announced in class and in the virtual learning environment.	
Collegerama	Yes No	

CME4601	Leadership and Strategic Management	5
Responsible Instructor	Prof.dr. P.W.C. Chan	
Instructor	Dr. M. Leijten	
Instructor	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The climate emergency, rapid urbanisation and resultant need to provide more affordable housing quickly, along with the urgency to transform our food, water, energy and waste infrastructure are key societal and sustainability transitions of our time. Furthermore, technological advancements in the digital space and the changing demographics are also demanding a rethink in the ways organisations function. Moreover, successive crises such as the Global Financial Crisis in the late 2000s and the more recent Covid-19 Global Pandemic have triggered calls for radical change. Put another way, if we keep doing what we have always done, then we will always get the (unsustainable) results that we have always gotten. Thus, there is a need to do things differently, and this is where leadership and strategic management can play a significant role. In tackling the societal and sustainability transitions highlighted above, simply relying on technical knowledge is not enough. These transition challenges are wicked problems, where simple solutions and incremental change are not sufficient. To address these transitions, there is also a need to inspire people from different backgrounds to activate radical change in our actions. Leadership skills are therefore critical to influence others to take the path less travelled. In addition, projects are increasingly considered and commissioned as drivers of change. Therefore, leadership is deeply connected to the delivery of projects. The key topics of this course include: a. Context: Understanding the context of leadership and strategy in transitions; b. Leadership theories: Complementing management and leadership, and shifting the mindset from planning and control (outputs) to influencing change and transformation (outcomes); c. Strategic management: Shifting the framing from strategy formulation and (long-term) planning to purpose-driven implementation and strategy-as-practice; d. Creativity and innovation: Shifting towards more open forms of innovation, and the role leadership and organisational strategy can play; e. Communication and negotiation: Understanding the power of communication and negotiation practices to influence outcomes; f. Values and impacts: Critically reflecting on the value, values and impacts of leadership and strategic management.	
Study Goals	The aim of this course is to enable students to develop their leadership skills and knowledge about strategic management of engineering projects. The intended learning outcomes include: (a) to develop critical awareness of leadership theories and apply these appropriately in analysing their personal leadership preferences; (b) to explain the significance and process of leadership in influencing others to achieve beneficial outcomes in the management of projects; (c) to develop critical awareness and knowledge of the strategic role that project managers and project management play in supporting wider organisational change outside of the boundaries of the project organisation, and; (d) to develop knowledge of the management of projects as a value creating process that translates policies and strategies into practice.	
Education Method	Lectures, Discussions, Group and Individual Assignments	
Assessment	It is important that the teaching, learning and assessment strategy takes into account that leadership development requires the individual to critically reflect in practice. The students will be required to submit two written assignments, including (1) a group assignment that combines leadership and strategic management in a change context; (2) a critical review on leadership and strategic management, and; (3) an individual reflection of personal leadership as applied to the management of projects.	
Expected prior Knowledge	Not applicable	
Academic Skills	Critical Thinking, Reflective Practice, Communication (Written and Oral Presentation), and Working in Teams.	
Literature & Study Materials	There will be directed reading provided each week to cover different aspects of leadership and strategy management in relation to managing projects. The course is also connected with the urgent need to drive societal and sustainability transitions. The following report commissioned by the Association for Project Management provides the contextual background for this course. Morris, P. W. G. (2017) Climate Change And What The Project Management Profession Should Be Doing About It: A UK Perspective, Princes Risborough: Association for Project Management, https://www.apm.org.uk/media/7496/climate-change-report.pdf.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Must choose at least one, but may also choose multiple

CME4800	Procurement of Complex Public Projects	5
Responsible Instructor	Dr.ir. A. Straub	
Responsible Instructor	Ir. J.H. Veldhuis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course provides an overview of the basic principles of value creation in public-private collaboration, and the design of collaborative networks and structures and related procurement strategies on portfolio and project level by public clients. Insight is given in the roles of public clients on construction markets, what their values are, and how they address (grand) challenges in asset management and construction in collaboration with private actors. By explaining and exploring different delivery methods and related procurement procedures and contract models, understanding is provided of the impact of these choices on the client organisation and the client - contractor relationship. Based on the case of a public client, all elements of the decision making around a public procurement process for construction activity are being discussed in their context: characteristics of the public portfolio and required construction activities, public values and societal challenges, characteristics of public and private organisations and their business models, stakeholders, supply market, delivery methods, distribution of tasks and responsibilities, risk allocation, the basics of asset management, contract models, procurement procedures and remuneration principles.	
Study Goals	At the end of this course you should be able to: LO1: explain the role of public clients and public values in construction; LO2: analyse the context of procuring construction activity in a public context; LO3: critically reflect on the application of delivery methods; LO4: design a procurement strategy; LO5: writing a scientifically based advisory report using scientific literature.	
Education Method	The course includes mandatory theoretical lectures, guest lectures and group tutorials. The course is structured around a real case the assessments are based upon.	
Assessment	Presentations and written assignments.	
Enrolment / Application	Brightspace	
Expected prior Knowledge	Interest in public clients and values, and procurement strategies for public projects in the built environment.	
Academic Skills	Students are expected to critically reflect on lectures and provided literature and to apply this to a case by writing a scientific advisory report. Students have to collaborate with other students to deliver group assignments.	
Literature & Study Materials	Case documents, (obligatory) scientific literature and presentations are made available via Brightspace.	
Judgement	Evaluation is based on group assignments (30%) and an individual assignment (70%). Both assignments have to be graded minimum 5,8 to pass the course. It is not possible only do the group assignment(s) or the individual assignment and doing the other assignment(s) the following year.	
Permitted Materials during Exam	Not Applicable	
Collegerama	No	

CME4850	Forms of Collaboration in Civil Engineering	5
Responsible Instructor	Ir. L.P.I.M. Hombergen	
Instructor	Prof.dr.ir. M.J.C.M. Hertogh	
Contact Hours / Week x/x/x/x	2 x 2hrs lectures per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Note: This course is the follow-up course for CIE5981. The course is slightly changed and extended with more material. In course CME4850 a review is given of the most common forms of collaboration in realising a project in civil engineering. The course discusses the state of the art. This is done in a fundamental way but the theory is clarified by means of practical examples illustrated by visiting lecturers. The course aims at preparing students fundamentally for the various forms of collaboration they will engage during his professional career. However it is emphasized that little attention will be paid to the literal contents of the various contracts. It is a matter of insight so that later on correct choices can be made for the appropriate form of contract for a specific type of project. The course is a synthesis of aspects in procurement, contracting and contractmanagement which are partly also given in other courses. The course is both an introduction to the business of building projects for more technically oriented students as a course for more building management oriented students with practical references. The following subjects will be dealt with: The principles of a governmental Tender system principles of an agreement and the elements that play a role in collaboration are discussed the control and contractmanagement of a project in relation to collaboration forms the contract and the corresponding components such as tasks, responsibilities and authorities the systems of reimbursement as a function of contract form risks, risk distribution, risk management, in various contract forms the family of design & construct, DBM, DBFM, DBFM, partnering, alliances, public private partnership, two phase contracts. practical examples illustrated by visiting lecturers international forms of collaboration like FIDIC and NEC4 the selection and choice of a contract partner new development in different countries. Summary The building industry is mostly project based. This often causes supply chains to be organised project by project. Parties in the building industry can choose between various forms of collaboration and contract models. Nowadays sometimes also portfolio's, group of projects, are tendered together. The different forms of collaborations are derived from the economic concepts such as value, price and cost. The course has a theoretical character but its contents is illustrated by means of practical examples by guest lecturers. The following forms of collaboration are dealt with: traditional contract, building team, general contracting and the families of Design & Construct. Discussed will be the contract-content with tasks, obligations, authorities, responsibilities, liabilities, systems of reimbursement and risk division. The following models of contract are discussed: design & construct, partnering, alliances, 2-phase contracts, public private partnership, risk management, risk sharing and contracts that deal with the organization during building.	
Study Goals	The students knowledge and skill in the following activities will be increased relative to the intellectual development level attained during his or her BSc study: - The understanding of the principles of project agreements; - The understanding and choice of forms of collaboration; - The evaluation of alternative forms of contract; - The various types of reimbursement; - The selection and choice of contract partners; - The understanding of the various components of contracts; - International contract types.	
Education Method	Lectures and discussions / Q&A	
Assessment	A written, eventually Multiple Choice, exam. Insight and overview will be tested to assess real life project situations.	
Exam Hours	The exam will take 2,5 hrs	
Expected prior Knowledge	Management of Engineering Projects / People are Key Edited by Hans L.M. Bakker and Jaap P. de Kleijn Chapter 9: Contracting. General academic knowledge and insights	
Academic Skills	Ability to analyse some different business concepts	

Literature & Study Materials	Obligatory lecture note(s)/textbook(s): Project and People / Mastering Success Chapter 12 Collaborative contracting in projects Reader: "Forms of collaboration in civil engineering" Available at Brightspace Material, papers and presentations made available on BrightSpace during the course/
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Dictionary
Collegerama	No

TPM024A	Methods for Risk Analysis and Management	5
Module Manager	Dr. M. Yang	
Instructor	Dr. M. Yang	
Responsible for assignments	Dr. M. Yang	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course consists of three blocks following a risk analytical approach: (A) introduction, (B) analysis, (C) assessment, treatment and follow-up. In block A Students will get familiar with the concepts and definition used in safety science and the basic steps of a risk analysis. In block B the course will focus on identification of hazards and qualitative/semi quantitative risk analysis. A selection of methods is explored to get acquainted with its application and limitations. For instance: Fault tree analysis, Event tree analysis, Bowtie diagram, Reliability Block Diagram, Markov Chain Analysis, Bayesian networks, Decision making under uncertainty. In block C the assessment of the outcomes of a risk analysis are put central. This includes acceptance, stakeholder perspectives, decision making and monitoring.	
Study Goals	After taking this course the student will have knowledge about: * the basic concepts of safety science and risk analysis * risk acceptance and decision making about risks After following this course, the student will be able to: * apply structured methods for risk identification, risk assessment, risk evaluation, and risk treatment.	
Education Method	The following methods may be used: Lectures, exercises and workgroup assignments. Students are expected to form groups to work on the projects applying risk-based methods in design or operation of socio-technical systems.	
Literature and Study Materials	The following textbook will be partially used: Engineering Risk Management (Meyer and Reniers, 2016; De Gruyter Publishing) Other required and additional material will be distributed via Brightspace	
Assessment	The primary assessment item of the course is a group project with an expected group size of 5-6 (depends on the class size). Assessment will be carried out on basis of groups at different stages of the project period. It will consist of three assignments, one presentation, and one final report. In total, there are 5 summative assessments. The grade distribution is: assignment 1 (proposal) - 15%, assignment 2 (methodology) -15%, assignment 3 (case study and results) - 15%, assignment 4 (oral presentation) - 25%, and assignment 5 (final report) - 30%. The minimum grade for each assessment item is 6,0. Feedback on the graded assignments can be reviewed by students upon appointment with the teach staff of this course within a week after the release of grades. The students are allowed one re-sit per assignment. It is not allowed to request a re-sit for an assignment for which students have received a pass (6,0 or higher). The re-sit format needs to be discussed with the teaching staff in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit must be taken within the current study year.	
Elective	Yes	
Category	MSc level	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Design & Integration

CME4100	Process Management	5
Responsible Instructor	Dr. M. Leijten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Complex construction- and engineering-related processes require collaboration between a variety of actors within a complex and dynamic environment. An environment that is influenced by various social, political and technical factors. These actors may be related to a process in various ways, e.g. as decision-makers, regulators, risk-taking participants and stakeholders, with diverging interests, resources and attitudes towards the process, being affected by a process and influencing it in different ways and degrees. In addition to the widely used project management, process management is needed to understand and adequately deal with these characteristics. This course aims at providing students with insights, concepts and skills needed to understand the nature of interaction in a multi-actor system regarding the initiation and development of, and decision-making in uncertain and dynamic situations. Students will analyse, develop and apply strategies, tools and arrangements that are part of the process management approach. There will be special attention to transitions and adaptive systems in an infrastructure and construction environment, but the principles are more broadly applicable. There will be a special focus on transition management and adaptive management to address the main current societal challenges related to construction management and infrastructure systems.	
Study Goals	After successfully completing this course, students should: - Be able to identify and analyse the perceptions, interests, strategies and institutional conditions that shape and constrain the behaviour of stakeholders. - Understand the dynamics of multi-actor interaction processes in uncertain and highly politicised contexts. - Understand information exchange in complex processes. - Be able to cope with substantive and strategic uncertainties and to show this in description and application. - Have developed knowledge, skills and competences on how to manage divergent and conflicting interests of stakeholders and the ability to apply this in practice. - Have developed competences on arriving at broadly-shared solutions. - Understand the essentials of decision-making in multi-actor networks. - Understand and apply theories on transition management and adaptive systems. - Have acquired knowledge, skills and competences on arrangements and tools which support collaboration, and be able to apply them.	
Education Method	Teaching is done through live lectures and interactive sessions, as well as learning materials offered via the online learning environment. There is also a group assignment that is offered in the virtual learning environment. Lectures and interactive sessions will be offered either on campus or in the virtual learning environment, depending on possible constraints in effect. Recordings from previous years are available on Collegerama. Interactive sessions are not recorded.	
Assessment	A group assignment (pass/fail) and an individual written exam. Students will receive further information and guidance on the exam during the lecture series and in the virtual learning environment. There will be two exam opportunities every academic year. The mark for the individual written exam will become the final mark for the course, providing the candidate has received a "pass" for the group assignment. Both scores will remain valid after the end of the academic year. A sufficient final mark (5.8 or higher) is required to pass the course. General grading rules of Delft University of Technology apply. The exams will be organised either on campus or remotely, depending on possible constraints in effect.	
Elective	Yes	
Tags	Process	
Expected prior Knowledge	There are no mandatory entrance courses. Proficiency in the English language is necessary.	
Academic Skills	- Collaboration and communication in a group - Academic writing - Building an argumentation - Analysis of complex situations - Proper referencing	
Literature & Study Materials	Required literature will be announced and provided in or via the virtual learning environment.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	In the current set-up, the exam will be closed-book. This means that no materials other than writing materials are allowed. If this policy changes, it will be announced in class and in the virtual learning environment.	
Collegerama	Yes No	

CME4250	Transformative Design	10
Responsible Instructor	Dr. E.J. Houwing	
Contact Hours / Week x/x/x/x	4 x 2hrs lectures per week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this new elective course the interaction between the transformative design of the built environment and its accompanying social, economic, political, and environmental issues is at stake. Students of the Master CME learn the needs of design & integration by working together with students from the Master AUBS following the MBE, Landscape Architecture and Urbanism tracks (from the coupled course peri-urban transformation). They collaborate on the transformation of a peripheral urban area in the Netherlands. These types of regions have a complex multitude of functions like residential, commercial, agricultural which challenge any transformation by existing pressures like urban sprawl, fragmented agricultural or industrial activity, remnants of natural and landscape values. Working together in groups you will develop a master plan and help integrate a more detailed, imaginable transformation proposal based on the domain-specific inputs. Students confront spatial qualities, opportunities, limitations and challenges of the chosen area with societal needs, transition goals, ownership issues and stakeholder claims. You design and define proposals on different scales and complexities including area, portfolio and project level. You will apply development strategies, feasibility criteria and research by design activities. This course is organised as a multidisciplinary experience combining approaches, knowledge and skills from the viewpoints of management, urban design and landscape development and is therefore representative for area development practice in the real world. This collaboration provides insight in the complementarity of viewpoints, data, tools, vocabulary, methods and fun.	
Study Goals	1. Students can analyse the characteristics, conditions, restrictions, opportunities and potentialities of peri-urban areas making use of a theoretical framework (analysis) 2. Students can develop a tentative spatial perspective (eg spatial, social, economic, political, environmental) in a peri-urban context including primary notions of phasing: from initiative to implementation (masterplan 0.5) 3. Students can synthesise varied input into an integrated, substantiated master plan for a peri-urban transformation area (masterplan 1.0) 4. Students can reflect on the implications of integrating domain-specific approaches in the application of a masterplan for a transformative design in the built environment	
Education Method	In this course four main educational methods are used: (1) Lectures (2) Studio work in groups (3) Studio work individually (4) Plenary presentations and discussions	
Assessment	This course is built on the following assessment structure: Analysis assignment (group work) for two weeks. This results in the 'Atlas'. Masterplan (group work) for two weeks. This results in the Masterplan 0.5. Design Assignment (individual work) for two weeks. This results in the reflection report (part 1). Masterplan (group work) for two weeks. This results in the Masterplan 1.0 (with input from the individual work). Integration (individual work) for two weeks. This results in the reflection report (part 2).	
Expected prior Knowledge	This is a master course for both students from Architecture as well as for students from CME. Collaborations between both disciplines is what you will bring along. This course is open to all CME students.	
Academic Skills	This is a Master course.	
Literature & Study Materials	materials will be provided during the course	
Judgement	A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page.	
Permitted Materials during Exam	No materials is needed. Deliverables are produced during group work and individual reflections.	
Collegerama	No	

CME4400	Entrepreneurial Engineering	5
Responsible Instructor	Dr. D.F.J. Schraven	
Responsible for assignments	Ir. L.P.I.M. Hombergen	
Co-responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Co-responsible for assignments	Prof.dr.ir. F.E.H.M. Smulders	
Contact Hours / Week x/x/x/x	2 x 2hrs lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	CME students, the specialisation Design & Integration.	
Course Contents	The course deals with the entrepreneurial traits in the civil engineering domain. Students are introduced with innovation and sustainability transition theories, new economy concepts and business concepts. They are challenged by 2 assignments how to implement a technological innovation in the civil engineering market structure and processes. The course consists of two cumulative modules: Analysis module, in which you analyse the success of a retrospective innovation case as a group, informed by innovation and transition theories and tools as your basis and inspired by relevant guest lectures from innovation experts from the civil engineering practice. Development module, in which you create a business plan for a real-time technological innovation as a group, informed by business model innovation processes, transition and adaptation methods.	
Study Goals	Finishing the course, that students are able to: - Understand and analyse an innovation case, with innovation theory to identify the successes and failures on its adoption; - Understand specific market structures of the civil engineering domain, the relations between public and private roles in this domain, and the multifaceted reality thereof; - Understand the market processes, specifically the public tendering processes for innovation-oriented projects; - Analyse the technical aspects of a technology / innovation to identify the robustness and the alternative implementations thereof; - Analyse the operational context of the projected implementations of a technology; - Create the role of your business in the pursuit of implementing the technology / innovation; - Create an adaptive and embedding business plan (flexible and future oriented) for the case. - From the perspective of a client or an established market party: be able to identify, stimulate, guide and embrace innovative innovation concepts.	
Education Method	The course is an activity-based learning experience, which is accompanied by lectures with the intent to inform and inspire the student with theoretical and practical insights.	
Assessment	Both in the Analysis and Development module, the assignment will be prepared in groups in the form of a report.	
Expected prior Knowledge	No prior knowledge is required.	
Academic Skills	To critically analyse the success of a previous innovation and extract lessons from it, and to be able to develop a business plan and model for an innovation in a civil engineering context.	
Literature & Study Materials	A selection of Book Chapters, academic articles and case-based data.	
Judgement	The assignments make up 40% and 60% respectively for the final grade. The combined grade of the 2 group assignments should be a 6.0 or higher in order to pass. Also both assignments should minimally be a 5.0 to count.	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Ethics Courses Must choose at least 1 out of these 4 courses

TPM003A	Water Ethics	5
Module Manager	Prof.dr.mr.ir. N. Doorn	
Co-responsible for assignments	Dr.ir. U. Pesch	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Water is essential for life. While we can survive weeks without food, we cannot survive more than a couple of days without water. This course aims to provide an introduction to philosophical water ethics. It will present the main philosophical issues in the water domain in a practical way by situating these issues within real cases. Questions addressed will include: Is it fair to drink water, knowing that the production of one cup of coffee requires 140 liters of water on average? Should we prioritize between different water uses, like agriculture, navigation, and recreation? Should we continue building hard flood defenses at the expense of the environment? Should local citizens be given a voice in water-related decision-making? These and other questions will be linked to ongoing discussions in other disciplines within philosophy, most notably in ethics of technology, philosophy, and ethics of risks, environmental ethics, climate ethics, and global justice. This course will provide students with an understanding of approaches in water engineering and water policy. Students will get experience discussing and evaluating ethical considerations in water engineering and water policy. Topics covered include water and justice, water and economic valuation, water and human rights, water and responsibility, and water and engineering. This course links to, amongst others, the following Sustainable Development Goals (SDGs; https://sdgs.un.org/goals): SDG6 Clean water and sanitation: Ensure availability and sustainable management of water and sanitation for all; SDG11 Sustainable cities and communities: Make cities and human settlements inclusive, safe, resilient and sustainable; SDG13 Climate action: Take urgent action to combat climate change and its impacts; SDG14 Life below water: Conserve and sustainably use the oceans, seas and marine resources for sustainable development.	
Study Goals	At the end of this course you will be able to: Describe the worlds water challenges, also in view of climatic and geographic developments. Describe and analyse the different water uses and explain how they may lead to competing demands; Describe how the worlds water challenges pose questions of justice and apply different concepts of justice to concrete water problems; Recognize and critically reflect on the methodological and ethical aspects of economic valuation in the context of water; Describe and apply different legal frameworks and rights concepts to the context of water; Describe and apply different concepts of responsibility to the context of water; Recognize and analyse the value-ladenness of water engineering and design; Describe and analyse current water governance practices and identify opportunities to include ethical considerations in water governance; Write a well-argued essay on a topic related to water ethics.	
Education Method	The structure of classes will combine lectures with seminars and debates.	
Books	"Water ethics: An introduction", by Neelke Doorn. Publisher: Rowman & Littlefield. 2019. ISBN: 9781786609519 (Pb) / 9781786609526 (E-book).	
Assessment	This course has no formative assessments or exams. It will be assessed via written assignments. The assessment consists of three parts: - Case report (group assignment in groups of 2-4 students); - Reflection (individual assignment); - Essay (individual assignment). Students are not allowed to write the case report and the essay on the same topic. Calculation of the course grade: The course grade is calculated as follows: 0.45 * case report grade + 0.15 * reflection grade + 0.4 * essay grade. The final grade will be rounded to the nearest half-integer. Students with a final grade of 6.0 or higher have passed the course. Validity: Grades for the different parts are valid only in the academic year in which the student participates in the tutorial sessions. Resit: In case of a grade lower than 6.0, students are allowed to write a new case report on a different case or a new essay on a different topic. This needs to be completed in the quarter following the quarter in which the tutorial sessions take place. Feedback: Students receive feedback on their assignments via Brightspace. Students receive detailed feedback on the draft version of their essay. Note: The assessment form does not depend on whether the course will be delivered online or on campus.	
Enrolment / Application	There are two versions of the water ethics course: a 3 EC version (TPM002A) and a 5 EC version (TPM003A). Students enrolled in the 3 EC course follow exactly the same educational activities as the students enrolled in the 3 EC course, except that they don't write the additional essay that students enrolled in the 5 EC version of the course need to write as one of their final assignments. This essay is worth 2 EC. All students should enroll in the 5 EC version of the course, also the students who do not wish to be eligible for this additional 2 EC. In the first week of the course, students can indicate whether they wish to follow the 3 EC or the 5 EC version of the course. There is no separate Brightspace site for the 3 EC course (TPM002A). Please enroll for the course via the Brightspace site of TPM003A, also if you wish to follow the 3 EC version of the course. Enrollment is possible until 1 November of the academic year. On that day, the total number of available places is determined. After 1 November, enrollment takes places upon approval and may be denied if there are no more places available.	
Special Information	Attendance of the tutorial sessions is mandatory. Students are allowed to miss only one session and only if they have a valid reason.	
Remarks	For ethics teaching at TU Delft, check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Sustainability Water Engineering Water management	
Targetgroup	All students enrolled in a MSc-program at TU Delft with an interest in water ethics.	
Category	MSc level	

WM0312CIE	Philosophy, Technology Assessment and Ethics for CIE	4
Module Manager	A.R. Gammon	
Instructor	C. Clare Shelley-Egan	
Responsible for assignments	A.R. Gammon	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module explores philosophical and ethical questions in civil engineering, as well as providing a basic introduction into Technology Assessment.	
Study Goals	After having completed the course you: - know the basic philosophical issues in engineering knowledge; - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	Virtual learning environment, introductory lecture and small number of question hours; mandatory assignment (more information will be provided during the introductory lecture and/or one of the question hours)	
Literature and Study Materials	- Ibo van de Poel and Lambèr Royakkers (2011). Ethics, Technology, and Engineering: An Introduction. Wiley Blackwell, West Sussex (ISBN 978-1-4443-3094-6 (hardcover) / 978-1-4443-3095-3 (paperback). - Additional reading material - Virtual learning environment (Brightspace).	
Assessment	Exam (80%) + mandatory assignment (20%) To pass the course, students must earn a 6.0 or higher. Students can carry partial grades over into future years. Students can resit the exam for a higher grade even if their initial grade is a passing one. Students can re-submit the mandatory assignment only in the case of an incomplete or a failing grade, not to improve a passing grade.	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Tags	Ethics Philosophy	
Category	MSc niveau	

WM0329TU	Ethics and Engineering	6
Module Manager	M. Sand	
Instructor	A.R. Gammon	
Co-responsible for assignments	A.R. Gammon	
Contact Hours / Week x/x/x/x	4/0/0/0 or 0/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	The first part of this course completely coincides with the WM320TU course. The WM329TU course is the WM320TU course plus the writing of an essay. Please check the WM320TU Brightspace for all relevant information about lectures, work sessions, exam, and the essay. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are able to write an opinion article on ethical or societal aspects of technology, according to academic standards of writing - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and 5 interactive work sessions concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	50%: grade for the participation in the course WM0320TU, which consists of a written exam (60%, individual grade), presentation, and active participation during the working group sessions (40%, same grade for all members of a presentation group). In order to pass the WM0320 part of the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. These partial grades are only valid for 1 academic year, meaning that if one or both are not passed during resit, both assessments need to be retaken the following academic year. 50%: grade of the essay, which is written individually (all steps in the essay writing procedure need to be followed). The grade of the essay (50%) conjoined with the grade of the WM0320TU part (50%) must be higher than 5.75. Only if the threshold is not met will it be possible to revise and resubmit the essay. It is not possible to retake the essay in another semester.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the number of workgroups. Please enrol as soon as possible and in any case before the start of the course via Brightspace.	
Remarks	The course is run twice each year, starting in quarters 1 and 3. The first part of this course (concluded with the written test) coincides with the course WM0320TU.	
Elective	Yes	
Category	MSc niveau	

WM0376TU	Ethics of Technological Risks	5
Module Manager	Prof.dr. S. Roeser	
Instructor	K. Maheshwari	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk.	
Education Method	Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.	
Assessment	On campus written exam (60% of the final grade) and essay (40% of the final grade). The exam is an individual assignment, the essay is written in pairs of students. Students need to have a pass (5.75 or higher) for each individual grade in order to pass the course. There is a resit of the written exam in the 4th quarter (date to be announced). In case students did not pass the essay initially, they have the opportunity to make further revisions within ca 2 weeks, but then the maximum grade is a 6, given the extra time. Partial grades are valid for the current and the following academic year. In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam [There is also a 3 ECTS version of this course, i.e. WM0375TU; the only difference is that for the 3 ECTS version students do not write an essay.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

Writing Skills (optional, but at maximum only one of these courses can be chosen)

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure your thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. You may not submit texts that were previously submitted for any other course. Students who submit their weekly assignments late, repeatedly miss deadlines or lack commitment may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when their work shows sufficient progress and weekly deadlines and all other conditions are met.	
Special Information	Enrolment is through Brightspace.	
Remarks	The course will only be offered when there is sufficient demand. If there are not enough enrolments, the course may be cancelled.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM0201TU	Technical Writing	2
Module Manager	Drs. P. van den Hoven	
Responsible for assignments	Drs. P. van den Hoven	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	Dutch	
Elective	Yes	
Category	BSc niveau	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

Project Courses (optional, but at maximum only one of these courses can be chosen)

CME2100-11	Research Internship	10
Responsible Instructor	Dr. E.J. Houwing	
Contact Hours / Week x/x/x/x	n.a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This research internship focusses on practical work experience in day-to-day practice of civil engineering companies or institutes (contractors, consultancies, government, non-governmental organizations, etc.) in the Netherlands or abroad. As the title suggests: (a major part of) the practical work is dedicated to an individual (research) assignment, to be agreed upon with the internship coordinator and TUD subject supervisor, following the steps of the internship manual (see CME-organization on Brightspace).	
Study Goals	The main objectives are * to develop your general engineering skills, * to develop your research skills, * to learn how to apply your technological know-how, * to put into practice any social and communicative skills you might have, * to gain a more complete insight into your own particular preferences and capabilities.	
Education Method	Practical work experience for a period of 8-10 weeks, focused on a specific research assignment. Before the student starts the internship, a brief project plan should be written and approved by the CME internship-coordinator (see course manual for the specific steps to be followed).	
Literature and Study Materials	On the Brightspace CME page you will find a digital version of the current course manual. This manual contains (amongst others) the information about the formalities that need to be arranged before the start of your internship. Any questions: please ask Mrs. Sandra Schuchmann (secretary, room 6.49).	
Assessment	The Internship is evaluated based on a written report to be handed in within 2 weeks after finishing the internship. An oral evaluation will follow with the students, the TUD subject supervisor and the coordinator. Also an evaluation of the company supervisor is required, see the course manual (Appendices) for all required forms. The report can be rejected if the student has not met the requirements as mentioned in the overview, if the level of the reporting is insufficient or if the work performed does not meet the requirements. One opportunity for rewriting the report will be given. If a report still does not meet the requirements after it has been resubmitted, a new internship will have to be carried out. -- Your grade for the internship is based on a judgement of the report on: 1) General engineering skills, 2) Technical knowledge, 3) Social and communication skills / reporting style, 4) Insight into own aptitude and affinity. The grading form can be found in the internship manual.	
Remarks	Finding a suitable internship is an important part of the course. Students should make sure to start organizing their internship well in time, preferably around 3-6 months before the planned date of departure! For information about the 'steps to be taken' please refer to the course documents (Course manual on CME Brightspace page).	
Expected prior Knowledge	Before starting the internship, you have finished at least all your obligatory courses of the CME program, since you need to reflect on the link between theory & practice. Preferably you also passed your electives. Typically, the internship is one of the final courses to complete (apart from the graduation project).	
Academic Skills	Important academic skills of thinking, writing and reflecting will be developed in the internship. The report is to be written in English.	
Literature & Study Materials	The study material consists of the (digital) course manual which is provided via Brightspace, by Sandra Schuchmann or the internship coordinator (Erik-Jan Houwing). All literature you have studied in your CME program can be relevant for your internship project: this obviously depends on the specific topic you will choose.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

CME4040-10EC	Joint Interdisciplinary Internship	10
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Important: CME4040-10EC is the CME coursecode for the Joint Interdisciplinary Project (JIP). Normally, the JIP is worth 15 EC, but due to overlap in the learning objectives for the JIP and other courses in the CME programme, the JIP is worth 10 EC for CME students. The additional 5 EC, are awarded extracurricularly for successful completion of the project. More information on the JIP can be found under the main JIP coursecode IFM4040 and on the corresponding Brightspace page. The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman, VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (ILO:K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: &#149; Problem statement review &#150; the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. &#149; Midterm review &#150; The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review &#150; the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). &#149; Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added. The minimum grade for the summative assessment is => 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.	
Expected prior Knowledge	To enter JIP, a student should: - be a 2nd-year master student; - have completed the first year of the master's degree programme and will enrol in the second year of the master&#146;s degree programme at TU Delft. - be full time available (40 hours per week) in Q5;	

- meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV
Students will be selected by the JIP core staff based on motivation, study progress and recommendation.

No

CME4900	Open Design Contractor Project	10
Responsible Instructor	Dr.ir. R. Binnekamp	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This course is a multi-disciplinary project based on deepening or broadening some of the engineering & management system (EMS) concepts of their interest, see https://odesys.nl/ (the creative commons: MitC, 3C-Plan, Preferendus etc.). These EMS concepts are open design systems for decision support within Engineering Asset management or Construction Projects management or Service Operations management. Offshore and/or Civil Engineering students a EMS Crossover The goal of this crossover course for civil and/or offshore engineering students is to learn about engineering asset management systems which are a group of concepts/tools and methods to either support the primary in-service operations management processes or the construction projects management processes for new delivery of engineering assets. This allows CE/ODE students to broaden their horizon within an integrative systems engineering and management context that is connected to their specific engineering domain of interest. Rather than only focusing on what to engineer, this crossover focuses especially on how to manage the project and operations management processes. This crossover is preferably directly linked with an contractor context. CME students a preparatory R&D project The goal of this course for CME students is to carry out a mini grassroot R&D project by applying one of the known engineering asset management systems concepts out of other ODL courses. This allows CME students to deepen their existing EMS knowledge and thereby prepare well for the final MSc project. Students can choose between the following two options: Option 1: A Crossover or R&D project within an international contractor practice on developing a self-chosen EMS concepts for contractor projects and/or services management (CoI). This development project is then directly linked with the course concepts of CME 4120 Information Systems, CME 4481 Systems Engineering Design and CME4701 Contractor Management Systems. Option 2: A Crossover or R&D project within an international service provider (SPoI) on developing a self-chosen EMS concepts for engineering asset management quality of services provisioning. This development project is then directly linked with the course concepts of CME 4300 Engineering Asset Management.	
Study Goals	1. Development and reflective practicioning of engineering management systems concepts: Engineering Asset management or Construction Projects management or Service Operations management (i.e., strategic asset management plan/ project delivery planning/ in service operation plan) . 2. Familiarizing with integrative learning and intercultural aspects within an international contractor organization. 3. Further enhancement of practicioning with R&D concepts as a preparation for a full engineering construction or asset management oriented graduation thesis within a ODL setup. 4. Demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/ learning achievements in an original ODL project or crossover response.	
Education Method	The course is based on the Open Design Learning concept (ODLc), see https://www.open-design.school/ . For this course the ODL response is descibing the process and results of this Full-time project work.	
Assessment	The deliverable of this course is the so-called Open Design learning response. This ODL response is a (group) deliverable based on the self-chosen EMS concept of interest. The Open Design Learning (ODL) response is an original enabler demonstrating both the group and personal learning and development achievements. The ODL response illustrates how the EMS concept of interest has been linked and evaluated to the self-chosen CoI/ SPoI using a: 1) logical review and/or diagramming 2) computer model. All of these (incl. the open glass box model) should be presented in a self-chosen format such as report/ elaborative presentation/ digital audio or video files/ animation/ website. The ODL must contain a clear justification of the individual contributions of each group member. Each group member must also write an individual Comment in which he/she writes a collegial review of at least 1 concept of the ODL response for which he/she was not responsible. Good collegial Comments make use of specific ODL Commendation aspects (see next section). The Open Design Learning Commendation principle will be applied as a grading rubric for the ODL response. Both the CoI/SPoI content characteristics, and the students learning process are integrated within these commendation principles. We call it commendation because when we grade your response, we start from a grade of 10 and only deduct points if aspects are missing/only partially worked out. There are 5 commendation categories: Connect/ Construct /Conclude & Conspect /Convey /Convince	
Expected prior Knowledge	One of the following courses : CME 4300, CME4481, CME4701, CME5030 for E&S	
Academic Skills	Skills to: curiously connect to a committed system of interest; critically convince others with a compelling response; clearly convey a communicative response; creatively convert concepts into a comparative response; construct a properly compiled open glass box model; contextualize in a combined and open space; comprehensively conclude and cogently conspect your open ended solutions.	
Literature & Study Materials	Reference material of the following courses : CME 4300, CME4481, CME4701, CME5030 for E&S	

Judgement	After handing in the ODL response you will receive your grade using the ODL commendation principle. To pass the course your ODL grade should be higher than or equal to 6. After commending your ODL response one plenary open dialogue session (max. 3 hours, somewhere between week 12 and 14 to be announced) with one of the constructors will be scheduled. Only during this session(s) we can reflect on your ODL response: bear in mind, the outcome of this reflection can result in a lower/equal/higher grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

CME4950	CME Research and Development Project	10
Responsible Instructor	Ir. J.S. Hoving	
Instructor	Dr. M. Leijten	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	x/x/x/x variable	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The student will, individually or in a team of multiple students, work on a CME research or development project which is part of ongoing CME research. This can be in collaboration with and under supervision by any CME staff member, including PhD's and postdocs. The research project can also be conducted in an affiliated research group (e.g. Deltares, TNO, research laboratory) or in a company R&D department. The CME research or development project provides students with the opportunity to work in a research environment, to master research techniques and further develop scientific thinking, argumentation and skills. It is an excellent opportunity for students to find out whether they aspire a scientific career and/or PhD-position. The student has to set up an assessment committee which consists of at least two supervisors, where at least one of them should be a full, associated or assistant professor in one of the faculties involved in the CME Master programme, i.e. Architecture & the Built Environment (ABE), Technology, Policy & Management (TPM) or Civil Engineering & Geosciences (CEG). Additionally, the student must inform their specialisation coordinator of the topic of their CME R&D project and the composition of the assessment committee (note here that this does not mean that the specialisation coordinator must be part of the assessment committee). The student is responsible for the planning, management and execution of the CME R&D project, in coordination with the assessment committee. The format of the deliverables of the CME R&D Project is not fixed and are agreed upon in consultation with the assessment committee, but the CME R&D project is always concluded by a final session with the assessment committee, including a presentation of the R&D project and a Q&A-session. The assessment committee is responsible for the assessment and determines the grade only after consultation of all committee members. As the CME R&D project is worth 10 credits, the student is expected to spend approximately 280 hours, which equals 7 weeks of 40 hours if executed full-time. However, the CME R&D project may also be executed part-time and is not necessarily restricted to the regular education periods, pending the approval of the assessment committee.	
Study Goals	By completing the CME Research & Development Project, you are able to: - Familiarize yourself with fundamental research and developments related to the topic; - Delineate a research, development or design problem and formulate a methodology to address the research or development statement in a well-structured way; - Plan, execute and manage the project effectively within the available time and resources; - Demonstrate your understanding of the scientific, societal and ethical context of the CME R&D Project; - Verify, interpret, and critically reflect on obtained results; - Draw relevant conclusions and give useful recommendations; - Present the CME R&D Project in a well-structured manner;	
Education Method	Self-study; support is available from the supervisors, i.e. the members of the thesis committee.	
Assessment	The assessment is based on: - the deliverables of the CME R&D project agreed upon with the assessment committee, - presentation of the CME R&D project, and - the Q&A-session after the presentation.	
Expected prior Knowledge	Any relevant courses in your individual CME study program. Especially the courses on Ethics and Methodology are of specific relevance.	
Academic Skills	Academic thinking and argumentation demonstrated by unequivocally communicating conclusions and recommendations, by addressing the knowledge, theory and motives, as well as societal and ethical considerations that underlie these.	
Literature & Study Materials	Relevant literature and study materials regarding the R&D topic are to be collected by the student from the open literature and may consist of research reports, peer-reviewed articles, books, proceedings and presentations.	
Judgement	Judgement by the assessment committee.	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Thesis

CME5100	CME Master Thesis Preparation	5
Responsible Instructor	Ir. J.S. Hoving	
Course Coordinator	Dr.ir. G.A. van Nederveen	
Course Coordinator	Dr. D.F.J. Schraven	
Course Coordinator	M.G.C. Bosch-Rekvelde	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	All CME students of cohort 2020-2021 and after, in preparation of their Master thesis project.	
Summary	CME5100 Master Thesis preparation is concluded by approval of the thesis proposal and always succeeded by CME5200 CME Master Thesis (30 credits). The Master Thesis preparation is not bound to a certain period. Further information regarding the CME graduation process is given during Quarterly CME Master Thesis information sessions; it is highly recommended for CME students who are (close to) starting their graduation to attend these sessions. The CME Master Thesis information sessions will be included in the lecture schedule (MyTimetable) and will be announced through the CME Brightspace community. For any further questions please refer to the graduation domain coordinators.	
Course Contents	The students will prepare their Master thesis by executing the following steps: 1. Formulate a rough research, development or design proposal in the field of your Construction Management and Engineering graduation domain* (max. 1 A4). This proposal may be based on a) the ideas of an involved company, b) university research, or c) personal ideas. 2. Submit your rough proposal to your graduation domain coordinator and get approval to continue the thesis preparation. 3. Find a main thesis supervisor in consultation with the graduation coordinator. 4. Establish a thesis committee in consultation with the main thesis supervisor, consisting of at least the main supervisor and a chairman, and, if desired and not already involved, find a company/organization to execute the Master thesis with. 5. If you choose to graduate at a company, arrange a graduation agreement in which all conditions for the graduation are stipulated and agreed upon by the company, the student and TU Delft. Note that the rough proposal from step 1 must be included in the graduation agreement. The standard TU Delft graduation agreement, the corresponding procedure and its conditions are available on the Faculty CEG student portal. 6. Finalize your thesis proposal, i.e. a full research, development or design proposal, including a) introduction of the thesis topic, b) problem description and definition, c) thesis objective, d) methodology and e) planning of the Master thesis project. 7. Defend your thesis proposal at the Master thesis kick-off meeting and get approval to start the CME Master Thesis. *) The thesis topic must be chosen within the scope of your CME graduation domain, which is either Design & Integration (D&I), Projects & People (P&P) or Systems & Decision Support (SDS). Each graduation domain may have specific requirements regarding the thesis topic; please check the CME Brightspace community for the corresponding information.	
Study Goals	After completing the CME Master Thesis Preparation, you have shown that you are able to: 1. Independently formulate a research proposal in consultation and agreement with the thesis committee and company representatives (if applicable) 2. Choose an applicable theoretical and/or experimental methodology aimed at reaching the thesis objectives 3. Provide argumentation in support of the thesis proposal and reflect on the possible outcomes of the thesis	
Education Method	Self-study; support is available from the graduation coordinators and/or prospective thesis supervisors.	
Prerequisites	Students may start CME5100 Master Thesis Preparation only if all of the following requirements are met: a. The student has completed at least 60 credits worth of courses out of the minimum 120 credits required for their Master programme in accordance with their registered individual study planning (ISP). b. The student has completed either one of the following courses: > CIE4030 Methodology for Scientific Research (3 EC) > CME5020 Research & Development Methods for P&P/D&I (3 EC) > CME5030 Research & Development Methods for E&S (5 EC) > CME5021 Research & Development Methods for CME (5 EC) c. The student has completed either one of the following ethics courses: > TPM003A Water Ethics (5 EC) > WM0312CIE Philosophy, Technology Assessment and Ethics for CIE (4 EC) > WM0329TU Ethics and Engineering (6 EC) > WM0376TU Ethics of Technological Risk (5 EC) Admission to start the graduation and this course may be granted while still awaiting exam results to fulfil above requirements, but only by request and based on a clarification by the student.	
Assessment	The course CME5100 CME Master Thesis Preparation is concluded only when the following requirements are met: a. The composition of the graduation supervision and assessment committee, in short the graduation committee, is approved in accordance with article 23 of the Rules and Guidelines Board of Examiners (R&G). b. The graduation agreement, if relevant, is signed and agreed upon by the company, the student and TU Delft. c. The thesis proposal is successfully defended, i.e. approved by the graduation committee, after assessment of the Master thesis proposal and the presentation of the thesis proposal at the Master thesis kick-off meeting. Formal approval of the thesis proposal is given by submission of the form: 'CME-2: Graduation Kick-off Proposal and Committee Approval' available at the Faculty CEG student portal.	
Enrolment / Application	You must ask permission to start CME5100 Master Thesis Preparation, either using the form: CME-1: Admission to start Graduation Thesis available at the Faculty CEG student portal, or through MyCase, a digital platform for the graduation process, if available.	
Contact	Coordinators: dr.ir. Marian Bosch-Rekvelde Coordinator Projects & People m.g.c.bosch-rekvelde@tudelft.nl	

dr. Daan Schraven
Coordinator Design & Integration
d.f.j.schraven@tudelft.nl

dr.ir. Sander van Nederveen
Coordinator Systems & Decision Support
g.a.vannederveen@tudelft.nl

CME Secretary:
Mrs. Sandra Schuchmann
Building 23 (CiTG), room 6.49, tel: +31 (0)15 27 84774
s.c.m.schuchmann@tudelft.nl

Expected prior Knowledge	See prerequisites. However, all courses in your individual CME study program may be relevant. The courses on Ethics and R&D Methodology are of specific relevance for the process during the CME Master Thesis.
Academic Skills	Critical thinking and argumentation, Written and verbal communication, Planning and time Management.
Literature & Study Materials	See courses on Ethics and R&D Methods. Any literature or other material that is relevant for the topic of your CME Master Thesis
Judgement	Judgement by the thesis committee.
Permitted Materials during Exam	Not applicable
Collegerama	No

CME5200	CME Master Thesis	30
Responsible Instructor	Ir. J.S. Hoving	
Course Coordinator	Dr.ir. G.A. van Nederveen	
Course Coordinator	M.G.C. Bosch-Rekvelde	
Course Coordinator	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Required for	All CME students of cohort 2020-2021 and after, as part of their graduation.	
Summary	CME5200 CME Master Thesis is always preceded by CME5100 Master Thesis preparation, which is concluded by formal approval of the thesis proposal. This approval can be registered either through submission of the form: 'CME-2: Graduation Kick-off Proposal and Committee approval' available at the Faculty CEG student portal, or through MyCase, a digital platform for the graduation process, when available. The CME Master Thesis is not bound to a certain period. Further information regarding the CME graduation process is given during Quarterly CME Master Thesis information sessions; it is highly recommended for CME students who are (close to) starting their graduation to attend these sessions. The CME Master Thesis information sessions will be included in the lecture schedule (MyTimetable) and will be announced through the CME Brightspace community. For any further questions please refer to the graduation domain coordinators.	
Course Contents	The CME Master Thesis consists of individually conducting a research, development or design project in the field of your Construction Management and Engineering graduation domain*. The thesis project is based on the thesis proposal approved by your thesis committee as part of CME5100 Master Thesis preparation. During the CME Master Thesis, you are to demonstrate the capacity to conduct your research, development or design project at the academic Master level by showing the ability to be able to independently plan, execute and manage your thesis project. During the thesis project you are additionally expected to demonstrate the ability to apply relevant theory, to set-up and execute a sound methodology, to critically interpret and reflect on obtained results and to present your findings in a well-structured manner through written and oral communications in proper English. As the CME Master Thesis is worth 30 credits, you are expected to finalise the thesis project in approximately 6 months. This excludes the time spent on CME5100 Master Thesis preparation. *) The thesis topic must be chosen within the scope of your CME graduation domain, which is either Design & Integration (D&I), Projects & People (P&P) or Systems & Decision Support (E&S). Each graduation domain may have specific requirements regarding the thesis topic; please check the CME community Brightspace for the corresponding information.	
Study Goals	After completing the CME Master Thesis, you have shown that you are able to: 1. Summarize relevant state-of-the-art research and developments and relate this to the thesis topic; 2. Demonstrate the ability to independently determine and apply relevant theory to the construction industry; 3. Manage the thesis process effectively considering the available time and resources; 4. Execute an applicable methodology aimed at reaching the thesis objectives 5. Effectively collaborate and communicate professionally and in an ethically and morally responsible manner 6. Verify, interpret, and reflect on obtained results in a critical manner; 7. Critically reflect on the thesis project at the level of your peers; 8. Present the thesis project in a well-structured written report or paper 9. Give a well-structured oral presentation regarding the thesis project to a wide audience that next to your peers, may include audience not directly familiar with the thesis topic; 10. Defend your work, results and reflections in presence of your peers by placing the thesis project into its scientific context and by explaining its contribution to the construction industry;	
Education Method	Self-study; support is available from the main supervisor and the other members of the thesis committee.	
Practical Guide	During CME5200 CME Master Thesis there will be several meetings with the full graduation committee present, including one or multiple progress meetings, the green-light meeting and the final graduation session. You are expected to organize these meetings in consultation with, and only after the agreement of, your university supervisor, who throughout the Master thesis is your first point of contact. For each meeting, it is your responsibility to make sure that all members of the graduation committee are informed about the date, time and location of the upcoming meeting. Any reservations of lecture/meeting rooms are to be made by the CME secretary Mrs. Sandra Schuchmann (see contact info). Progress Meetings: Not later than 1 week prior to each progress meeting, you are requested to send a digital version of the draft report to all members of the graduation committee. During the progress meetings, you are expected to present your progress and to discuss the way forward. Green-light Meeting: Before the green-light meeting can take place, you must have completed all courses, other than CME5200 CME Master Thesis, as listed in your registered individual study planning (ISP). Not later than 1 week prior to this meeting, you are requested to send a final draft of the thesis to all committee members. During the green-light meeting, you are expected to give a 20- to 25-minute presentation (excl. Q&A) giving a full overview of the graduation project with a focus on results and conclusions. Based on the thesis draft and the presentation, the graduation committee determines whether you are ready to organise the final graduation session. Approval for this, commonly known as the green-light, is only given if the graduation committee concludes that the quality of the work done as presented and defended by the student during the green-light meeting, is sufficient to graduate (with a grade of at least 6.0). Therefore, at the green-light meeting at least two examiners (as appointed by the Board of Examiners) must be present. If the green-light is not given, the committee informs you whether, and how, approval can be obtained. If the green-light is given, you are to submit the form CME-3: Green Light for Graduation Thesis Defence, available at the Faculty CEG student portal, to the student administration no later than 20 working days before the graduation date (see R&G Board of Examiners, article 9b). Thus, the green-light meeting must always take place at least 4 weeks before the final graduation session. During this time, the administration office checks whether you meet all the program requirements, including successful completion of your registered individual CME study program (ISP).	

	Final Graduation Session: Not later than 1 week prior to the final graduation session, you are to submit a digital version of the final Master thesis to all members of the graduation committee. The final thesis is an adaptation of the final draft (delivered prior to the Green-light meeting) that includes corrections and improvements according to the agreements made during the green light meeting. During the final graduation session, you are to give a public presentation focusing on your objectives, obtained results, conclusions and own contribution. In addition, the presentation should be inspiring and understandable for the general public. After the presentation, you are to publicly defend your work by answering the questions and challenges posed by your thesis committee and possibly by the audience, as well as critically reflect on your own work. If the combination of the thesis, presentation and defence are deemed worthy by the thesis committee, the session commences with the diploma ceremony. Students may start CME5200 CME Master Thesis when they have successfully completed the course CME5100 CME Master Thesis Preparation. The requirements for the course CME5100 Master Thesis Preparation, and thereby the requirements to start the graduation, are given in the course description for CME5100. Note that the course CME5100 CME Master Thesis Preparation is concluded only when the following requirements are met: a. The composition of the graduation supervision and assessment committee, in short the graduation committee, is approved in accordance with article 23 of the Rules and Guidelines Board of Examiners (R&G). b. The graduation agreement, if relevant, is signed and agreed upon by the company, the student and TU Delft. c. The thesis proposal is successfully defended, i.e. approved by the graduation committee, after assessment of the Master thesis proposal and the presentation of the thesis proposal at the Master thesis kick-off meeting. Formal approval of the thesis proposal is either given by submission of the form 'CME-2: Graduation Kick-off Proposal and Committee Approval' available at the Faculty CEG student portal, or through MyCase, a digital platform for the graduation process, when available.. This is the starting point of your CME Master Thesis.
Prerequisites	
Assessment	At the final graduation session, the members of the graduation committee together determine the CME5200 CME Master Thesis grade during a closed meeting and without presence of the student. This meeting is part of the final graduation session and succeeds both the final presentation and the Q&A session. The assessment of the Master Thesis is performed by the examiner members of the thesis committee taking into account the final attainments of the CME program and using the rubric for CME5200 CME Master Thesis. University members that are not appointed as examiner by the Board of Examiners and external (company) members only have an advisory role. If the examiner members cannot find consensus on the grade, the committee chair determines the grade based on the input of the other members of the graduation committee. The CME5200 CME Master Thesis rubric distinguishes the following three main aspects: 1. Content/Product: Quality of research, development or design presented in the Master Thesis; This aspect has to be sufficient in itself to lead to a sufficient overall score. 2. Process: Working and learning attitude during the thesis project. 3. Presentation and Communication: a) Report, b) Oral presentation and Defence
Enrolment / Application	To start the CME Master Thesis students must have their thesis proposal approved and successfully completed CME5100 Master Thesis Preparation. Formal approval of the thesis proposal and thereby permission to start the CME Master Thesis is either given by submission of the form: 'CME-2: Graduation Kick-off Proposal and Committee Approval' available at the Faculty CEG student portal, or through MyCase, a digital platform for the graduation process, if available. The following graduation forms exist and are available at the TU Delft student portal and at the CME-community on Brightspace: CME-1: Admission to start Graduation Thesis CME-2: Graduation Kick-off Proposal and Committee Approval CME-2a: Changes to Composition of the Graduation Committee CME-3: Green Light for Graduation Thesis Defence
Special Information	Composition, roles and presence of the graduation committee: The composition of the graduation committee must be in accordance with article 23 of the Rules and Guidelines Board of Examiners (R&G). Your university supervisor is your first point of contact throughout the Master thesis. The chair of the graduation committee is responsible for ensuring that the relevant rules regarding CME5200 CME Master Thesis are followed. If a member of the graduation is not able to be present (online or in person) at either the green-light meeting or the final graduation session, note that the green-light meeting and the final graduation session may only take place if at least two examiners (as appointed by the Board of Examiners) are present. Repository and Confidentiality: In preparation of the final graduation session students must upload their Master Thesis to the TU Delft Education Repository. In consultation with the relevant parties involved, a redacted version of the thesis may be uploaded or a temporary embargo may be enacted. An embargo can be placed on a confidential thesis for up to two years, with a possible maximum extension of three years. Only after the student has fulfilled the obligation as described in this section, the final grade for CME5200 CME Master Thesis will be registered. For any procedures related to the graduation agreement, confidentiality and embargos please contact Legal Services. For contact info and involved procedures see the Forms Master page at the CEG Student Portal. For additional information about the CME Master Thesis, its process and any possible additional requirements per graduation domain please check the CME-community on Brightspace.
Contact	Coordinators: dr.ir. Marian Bosch-Rekveldt Coordinator Projects & People m.g.c.bosch-rekveldt@tudelft.nl dr. Daan Schraven Coordinator Design & Integration d.f.j.schraven@tudelft.nl dr.ir. Sander van Nederveen Coordinator Systems & Decision Support g.a.vannederveen@tudelft.nl CME Secretary: Mrs. Sandra Schuchmann Building 23 (CiTG), room 6.49, tel: +31 (0)15 27 84774 s.c.m.schuchmann@tudelft.nl
Expected prior Knowledge	All courses in your individual CME study program. Especially the courses on Ethics and Methodology are of specific relevance for the process during the CME Master Thesis.
Academic Skills	Critical thinking and argumentation, written and verbal communication, planning and time management.

Literature & Study Materials	Relevant literature and study materials regarding the thesis topic are to be collected by the student from the open literature and may consist of research reports, peer-reviewed articles, books and proceedings.
Judgement	Judgement by the thesis committee.
Permitted Materials during Exam	Not applicable.
Collegerama	No

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	Master Construction Management and Engineering

CME Thesis for students who started before September 2020

CME2000	Master Thesis	32
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Course Language	English	

CME2001	Master Thesis, Preparation	4
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Course Language	English	

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